

2

Acids and Bases; Functional Groups

Goals for Chapter 2

1 Identify the molecular features that cause compounds to be polar and to engage in hydrogen bonding.

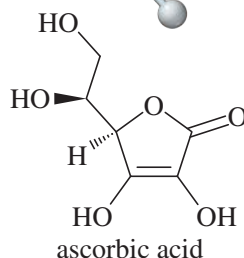
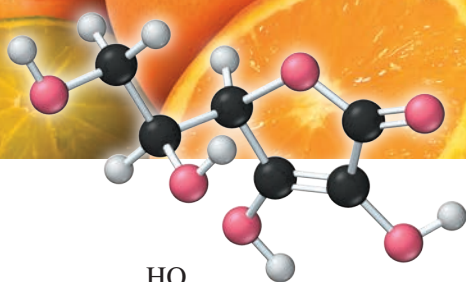
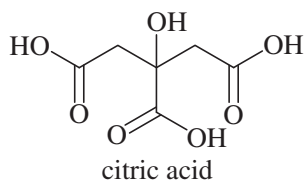
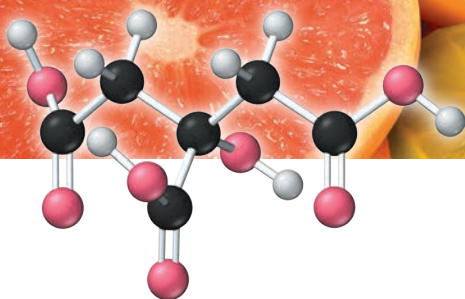
2 Predict general trends in physical properties such as boiling points and solubilities.

3 Identify acids, bases, electrophiles, and nucleophiles. Compare their strengths and predict their reactions based on structure and bonding, as well as K_a and pK_a values.

4 Identify the nucleophiles and electrophiles in Lewis acid–base reactions and use curved arrows to show the flow of electrons.

5 Identify the general classes of organic compounds.

◀ Citric acid and ascorbic acid (vitamin C) are two water-soluble organic acids found in citrus fruits. Citric acid plays a central role in metabolism, and ascorbic acid is needed for growth and repair of collagen, a structural protein in connective tissue. Unlike most animals, humans cannot synthesize their own ascorbic acid. Ascorbic acid deficiency causes scurvy, with debilitating and often deadly symptoms resulting from damage to connective tissues.



Chemists categorize organic compounds into families based on characteristic groups of atoms called **functional groups**. These functional groups determine the properties of their parent compounds, and they serve as acidic or basic sites within the molecules. The acidic and basic properties of organic compounds are crucial to understanding their stability and reactivity. Most organic compounds can react as both acids in a basic environment and bases in an acidic environment. Nearly all the substances around you and inside you are acids and bases—from the food you eat to the products you use to clean up afterward.

TABLE 2-1 Bond Dipole Moments (Debye) for Some Common Covalent Bonds

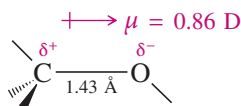
Bond	Dipole Moment, μ	Bond	Dipole Moment, μ
$\begin{array}{c} \text{+} \rightarrow \\ \text{C} \rightarrow \text{N} \end{array}$	0.22 D	$\begin{array}{c} \text{+} \rightarrow \\ \text{H} \rightarrow \text{C} \end{array}$	0.3 D
$\begin{array}{c} \text{+} \rightarrow \\ \text{C} \rightarrow \text{O} \end{array}$	0.86 D	$\begin{array}{c} \text{+} \rightarrow \\ \text{H} \rightarrow \text{N} \end{array}$	1.31 D
$\begin{array}{c} \text{+} \rightarrow \\ \text{C} \rightarrow \text{F} \end{array}$	1.51 D	$\begin{array}{c} \text{+} \rightarrow \\ \text{H} \rightarrow \text{O} \end{array}$	1.53 D
$\begin{array}{c} \text{+} \rightarrow \\ \text{C} \rightarrow \text{Cl} \end{array}$	1.56 D	$\begin{array}{c} \text{+} \rightarrow \\ \text{C} = \text{O} \end{array}$	2.4 D
$\begin{array}{c} \text{+} \rightarrow \\ \text{C} \rightarrow \text{Br} \end{array}$	1.48 D	$\begin{array}{c} \text{+} \rightarrow \\ \text{C} \equiv \text{N} \end{array}$	3.6 D
$\begin{array}{c} \text{+} \rightarrow \\ \text{C} \rightarrow \text{I} \end{array}$	1.29 D		

SOLVED PROBLEM 2-1

Calculate the amount of charge separation for a typical C—O single bond, with a bond length of 1.43 Å and a dipole moment of 0.86 D.

SOLUTION

The bond is polarized with a partial negative charge on oxygen and an equal amount of positive charge on carbon.



Using the formula for the dipole moment, we have

$$0.86 \text{ D} = 4.8 \times \delta \times 1.43 \text{ Å}$$

$$\delta = 0.125 e$$

The amount δ of charge separation is about 0.125 electronic charge, so the carbon atom has about an eighth of a positive charge, and the oxygen atom has about an eighth of a negative charge.

PROBLEM 2-1

The C=O double bond has a dipole moment of about 2.4 D and a bond length of about 1.23 Å.

- Calculate the amount of charge separation in this bond.
- Use this information to evaluate the relative importance of the following two resonance contributors:



(R is a general symbol for a carbon-containing group.)

2-1B Molecular Dipole Moments

A **molecular dipole moment** is the dipole moment of the molecule taken as a whole. It is a good indicator of a molecule's overall polarity. Molecular dipole moments can be measured directly, in contrast to bond dipole moments, which must be estimated by comparing various compounds. The value of the molecular dipole moment is equal to

the *vector* sum of the individual bond dipole moments. This vector sum reflects both the magnitude and the direction of each individual bond dipole moment.

For example, formaldehyde has one strongly polar C=O bond, and carbon dioxide has two. We might expect CO₂ to have the larger dipole moment, but its dipole moment is actually zero. The symmetry of the carbon dioxide molecule explains this surprising result. The structures of formaldehyde and carbon dioxide are shown here, together with their electrostatic potential maps. These electrostatic potential maps show the directions of the bond dipole moments, with red at the negative ends and blue at the positive ends of the dipoles. In carbon dioxide, the bond dipole moments are oriented in opposite directions, so they cancel each other.

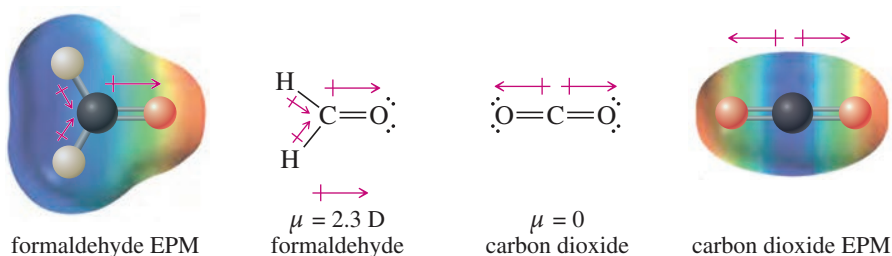


Figure 2-1 shows some examples of molecular dipole moments. Notice that the dipole moment of C—H bonds is small, so we often treat C—H bonds as nearly nonpolar. Also note that the tetrahedral symmetry of CCl₄ positions the four C—Cl dipole moments in directions so that they cancel. A partial canceling of the bond dipole moments explains why CHCl₃, with three C—Cl bonds, has a smaller molecular dipole moment than CH₃Cl, with only one.

Lone pairs of electrons contribute to the dipole moments of bonds and molecules. Each lone pair corresponds to a charge separation, with the nucleus having a partial positive charge balanced by the negative charge of the lone pair. Figure 2-2 shows four molecules with lone pairs and large dipole moments. Notice how the lone pairs contribute to the large dipole moments, especially in the C=O and C≡N bonds. Also notice the red areas in the electrostatic potential maps, indicating high negative potential in the electron-rich regions of the lone pairs.

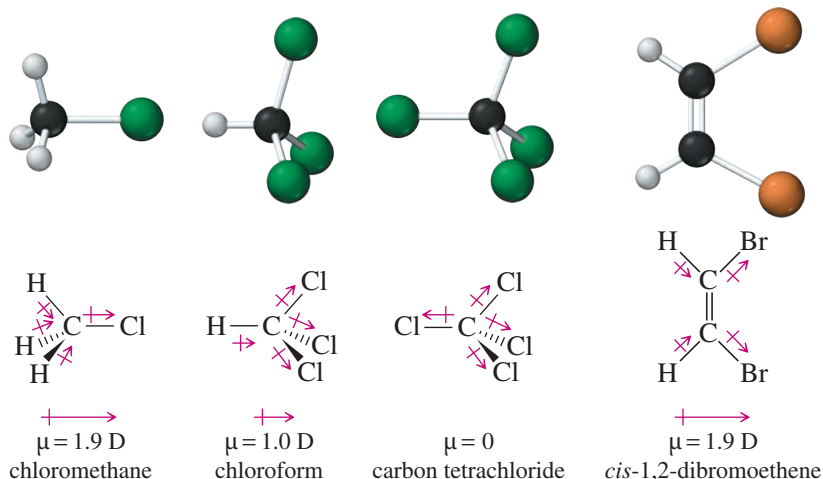


FIGURE 2-1 Molecular dipole moments. A molecular dipole moment is the vector sum of the individual bond dipole moments.

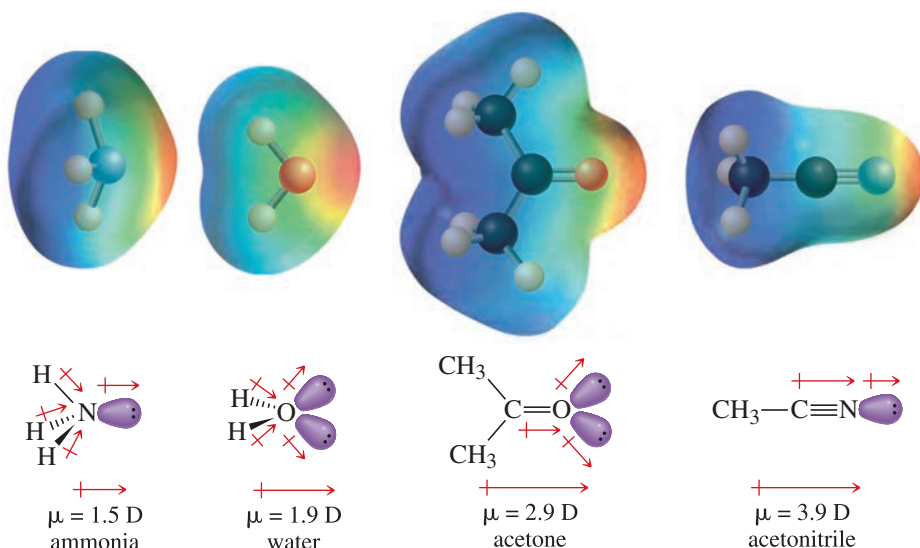


FIGURE 2-2 Effects of lone pairs on dipole moments. Lone pairs can make large contributions to molecular dipole moments.

PROBLEM 2-2

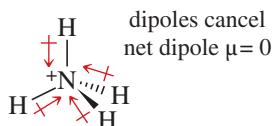
The N—F bond is more polar than the N—H bond, but NF_3 has a *smaller* dipole moment than NH_3 . Explain this curious result.



PROBLEM 2-3 (PARTIALLY SOLVED)

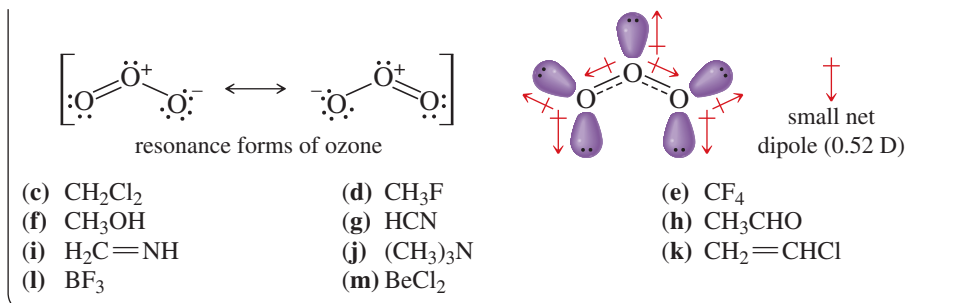
For each of the following compounds:

1. Draw the Lewis structure.
 2. Show how the bond dipole moments (and those of any nonbonding pairs of electrons) contribute to the molecular dipole moment.
 3. Estimate whether the compound will have a large, small, or zero dipole moment.
 - (a) NH_4^+
 - (b) O_3
- (a) NH_4^+ has four polar N—H bonds. These bonds are probably more polarized than a typical N—H bond, because the N in NH_4^+ bears a formal positive charge. Nevertheless, these four polar bonds have a symmetric tetrahedral arrangement so they cancel each other.

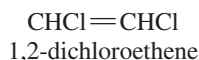


- (b) Ozone (O_3) is an sp^2 hybrid structure, with a lone pair on the central oxygen atom. Therefore, O_3 must be bent. The resonance structures imply partial negative charges on the outer oxygens and a partial positive charge on the central oxygen. The lone pair on the central oxygen cancels part, but not all, of the vector sum of the two O—O dipoles. The resulting net dipole is relatively small.

(continued)

**PROBLEM 2-4**

Two isomers of 1,2-dichloroethene are known. One has a dipole moment of 2.4 D; the other has zero dipole moment. Draw the two isomers, and explain why one has zero dipole moment.

**2-2 Intermolecular Forces**

When two molecules approach, they attract or repel each other. This interaction can be described fairly simply in the case of atoms (such as the noble gases) or simple molecules such as H_2 and Cl_2 . In general, the forces are attractive until the molecules come so close that they infringe on each other's van der Waals radius. When this happens, the small attractive force quickly becomes a large repulsive force, and the molecules "bounce" off each other. With complicated organic molecules, these attractive and repulsive forces are more difficult to predict. We can still describe the nature of the forces, however, and we can show how they affect the physical properties of organic compounds.

Attractions between molecules are particularly important in solids and liquids. In these "condensed" phases, the molecules are continuously in contact with each other. The melting points, boiling points, and solubilities of organic compounds show the effects of these forces. Boiling points roughly indicate the strength of intermolecular forces, because those forces must be overcome to boil the compound. Three major kinds of attractive forces cause molecules to associate into solids and liquids:

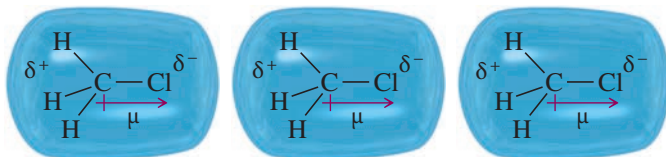
1. the dipole–dipole forces of polar molecules;
2. the London dispersion forces that affect all molecules; and
3. the "hydrogen bonds" that link molecules having —OH or —NH groups.

2-2A Dipole–Dipole Forces

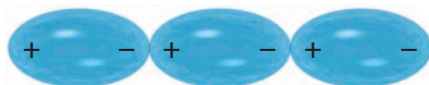
Most molecules have permanent dipole moments as a result of their polar bonds. Each molecular dipole moment has a positive end and a negative end. The most stable arrangement has the positive end of one dipole close to the negative end of another. When two negative ends or two positive ends approach each other, they repel, but they may turn and orient themselves in the more stable positive-to-negative arrangement. **Dipole–dipole forces**, therefore, are generally attractive intermolecular forces resulting from the attraction of the positive and negative ends of the dipole moments of polar molecules. Figure 2-3 shows the attractive and repulsive orientations of polar molecules, using chloromethane as the example.

Polar molecules are mostly oriented in the lower-energy positive-to-negative arrangement, and the net force is attractive. This attraction must be overcome when the liquid vaporizes, resulting in larger heats of vaporization and higher boiling points for strongly polar compounds.

attraction (common)



symbolized by



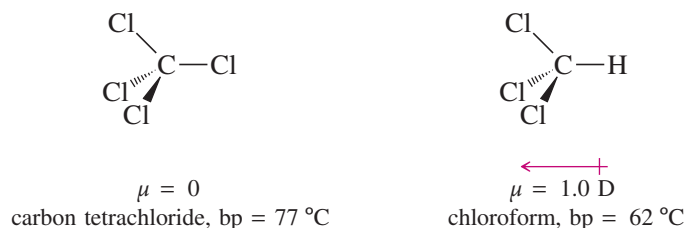
repulsion (uncommon)



FIGURE 2-3 Dipole–dipole interactions. Dipole–dipole interactions result from the approach of two polar molecules. If their positive and negative ends approach, the interaction is attractive. If two negative ends or two positive ends approach, the interaction is repulsive. In a liquid or a solid, the molecules orient mostly with their positive and negative ends together, and the net force is attractive.

2-2B The London Dispersion Force

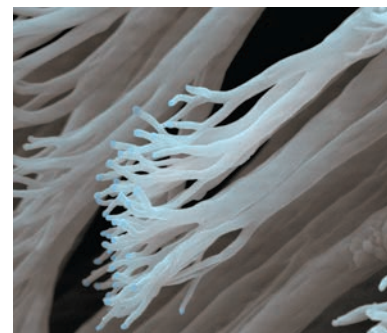
Carbon tetrachloride (CCl_4) has a zero dipole moment, yet its boiling point is higher than that of chloroform ($\mu = 1.0 \text{ D}$). Clearly, there must be some kind of force other than dipole–dipole forces holding the molecules of carbon tetrachloride together.



In nonpolar molecules such as carbon tetrachloride, the principal attractive force is the **London dispersion force**, one of the **van der Waals forces** (Figure 2-4). The London force arises from temporary dipole moments that are induced in a molecule by other nearby molecules. Even though carbon tetrachloride has no permanent dipole moment, the electrons are not always evenly distributed. A small temporary dipole moment is induced when one molecule approaches another molecule in which the electrons are slightly displaced from a symmetrical arrangement. The electrons in the approaching molecule are displaced slightly so that an attractive dipole–dipole interaction results.

These temporary dipoles last only a fraction of a second, and they constantly change; yet they are correlated so that their net force is attractive. This attractive force depends on close surface contact of two molecules, so it is roughly proportional to the molecular surface area. Carbon tetrachloride has a larger surface area than chloroform (a chlorine atom is much larger than a hydrogen atom), so the intermolecular London dispersion attractions between carbon tetrachloride molecules are stronger than they are between chloroform molecules.

We can see the effects of London forces in the boiling points of simple hydrocarbons. If we compare the boiling points of several isomers, the isomers with larger surface areas (and greater potential for London force attraction) have higher boiling points. The boiling points of three C_5H_{12} isomers are given here. The long-chain isomer (*n*-pentane) has the greatest surface area and the highest boiling point. As the amount of chain branching increases, the molecule becomes more spherical and its surface area decreases. The most highly branched isomer (neopentane) has the smallest surface area and the lowest boiling point.



A gecko easily climbs a polished glass window, even though the glass has no crevices. A gecko's toe has thousands of tiny hairs, with each hair split into hundreds of tiny tips. Each tiny hair tip is attracted to the surface by van der Waals forces. Millions of hair tips provide a large surface area, so these weak intermolecular attractions easily support the gecko's weight.

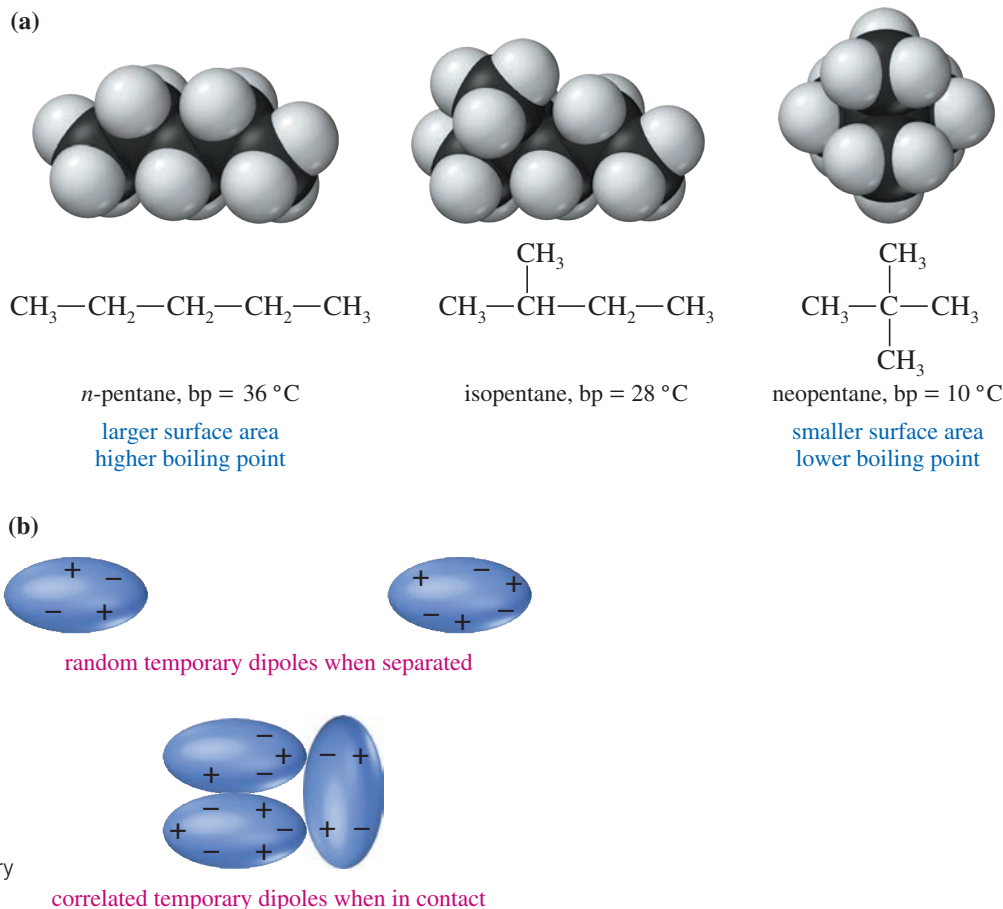


FIGURE 2-4 London dispersion forces between nonpolar molecules. London dispersion forces result from the attraction of correlated temporary dipoles.

Application: Hydrogen Bonding

When we wash clothes, water molecules insert themselves between the cellulose molecules and make them slide over each other, causing wrinkles. When we iron them, the combination of heat and moisture from the clothing iron breaks the hydrogen bonds and allows the cellulose molecules to lie parallel to one another, thereby smoothening the wrinkles in our clothes.

2-2C Hydrogen Bonding

A **hydrogen bond** is not a true bond but a particularly strong dipole–dipole attraction. A hydrogen atom can participate in hydrogen bonding if it is bonded to oxygen, nitrogen, or fluorine. Organic compounds do not contain H—F bonds, so we consider only N—H and O—H hydrogens to be hydrogen bonded (Figure 2-5).

The O—H and N—H bonds are strongly polarized, leaving the hydrogen atom with a partial positive charge. This electrophilic hydrogen has a strong affinity for nonbonding electrons, and it forms intermolecular attachments with the nonbonding electrons on oxygen or nitrogen atoms.

Hydrogen bonding is a strong form of intermolecular attraction, yet it is much weaker than a normal C—H, N—H, or O—H covalent bond. Breaking a hydrogen bond requires about 20 kJ/mol (5 kcal/mol), compared with about 400 kJ/mol (about 100 kcal/mol) required to break a C—H, N—H, or O—H bond.

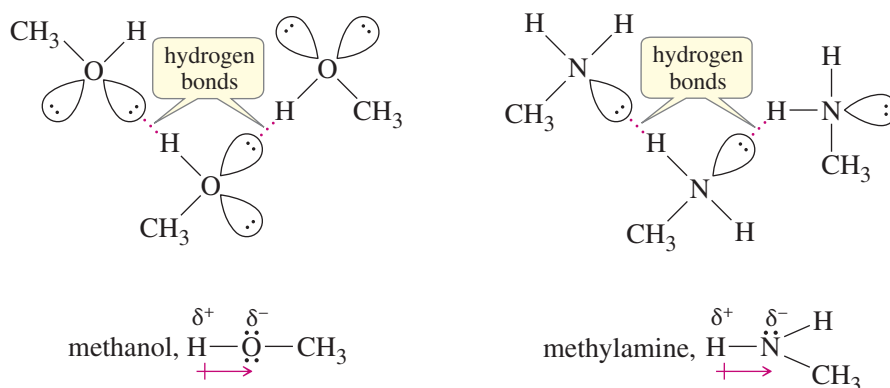
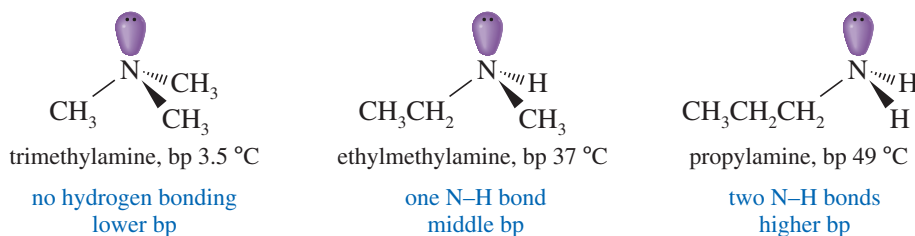
Hydrogen bonding has a large effect on the physical properties of organic compounds, as shown by the boiling points of ethanol (ethyl alcohol) and dimethyl ether, two isomers of molecular formula C₂H₆O:



These two isomers have the same size and the same molecular weight. Alcohols such as ethanol have O—H hydrogens, however, so they are extensively hydrogen bonded. Dimethyl ether has no O—H hydrogen, so it cannot form hydrogen

bonds. As a result of its hydrogen bonding, ethanol has a boiling point more than 100 °C higher than that of dimethyl ether. Both ethanol and dimethyl ether can form hydrogen bonds with the —OH groups of water, however, so both of them are soluble in water.

The effect of N—H hydrogen bonding on boiling points can be seen in the isomers of formula C_3H_9N shown below. Trimethylamine has no N—H hydrogens, so it is not hydrogen bonded. Ethylmethanamine has one N—H hydrogen atom, and the resulting hydrogen bonding raises its boiling point about 34 °C above that of trimethylamine. Propylamine, with two N—H hydrogens, is more extensively hydrogen bonded and has the highest boiling point of these three isomers.



Application: Energy Conversion

The joule is the SI unit for energy, corresponding to the energy of a mass of 1 kg moving at 1 meter per second. The calorie is the cgs unit for energy, corresponding to the energy required to raise the temperature of 1 gram of water from 14.5 °C to 15.5 °C. Both units are widely used. They are related by 1 cal = 4.184 J, or 1 kcal = 4.184 kJ.

FIGURE 2-5 Hydrogen bonding. Hydrogen bonding is a strong attraction between an electrophilic O—H or N—H hydrogen atom and a pair of nonbonding electrons.

Alcohols form stronger hydrogen bonds than amines because oxygen is more electronegative than nitrogen. Thus, the O—H bond is more strongly polarized than the N—H bond. This effect is seen in the boiling points of the preceding isomers, with more than 100 °C difference in the boiling points of ethanol and dimethyl ether, compared with a 34 °C difference for ethylmethanamine and trimethylamine.

PROBLEM 2-5

Draw the hydrogen bonding that takes place between

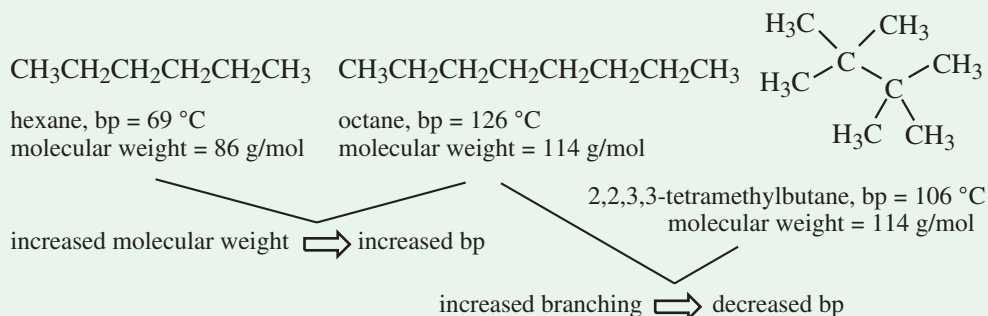
- two molecules of ethanol.
- two molecules of propylamine.
- a molecule of dimethyl ether and two molecules of water.
- two molecules of trimethylamine and a molecule of water.

FOCUS Predicting Relative Boiling Points

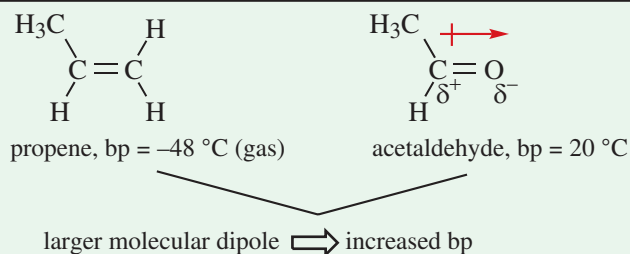
Boiling is the physical separation of individual molecules from the liquid phase to the vapor phase. In order for a liquid to boil, all *intermolecular forces* need to be overcome. We group these forces into three categories for organic molecules.

The London dispersion force

affects all atoms and molecules, and depends on the surface area of the molecule that contacts its neighbors. Larger molecules have greater intermolecular contact, so there is a rough correlation of higher boiling point with increasing molecular weight. In contrast, increasing branching decreases the surface area and decreases the boiling point.

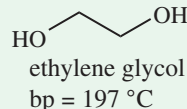
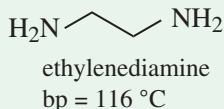


Dipole-dipole force occurs in molecules with electronegative atoms that have a permanent dipole. The stronger the dipole, the higher the boiling point.



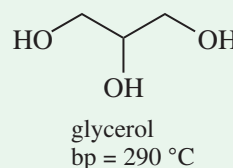
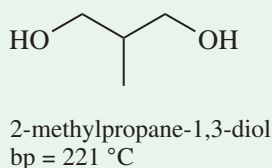
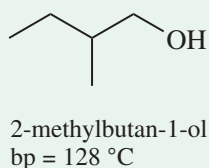
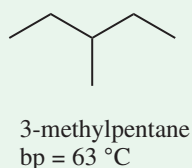
Hydrogen bonding is a special case of dipole-dipole force in which the molecule has an O—H or N—H group. In addition to the dipole-dipole force, the H on the O or N of one molecule forms a strong attraction, a **hydrogen bond**, to the O or N of another molecule. This makes it the strongest of the three intermolecular forces. Two generalizations apply:

- Molecules with OH groups have stronger hydrogen bonds, and therefore higher boiling points, than similar compounds with NH groups.



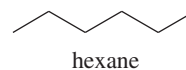
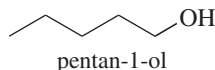
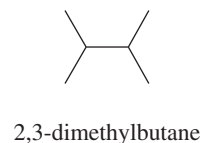
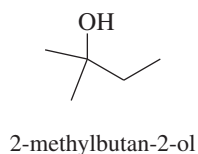
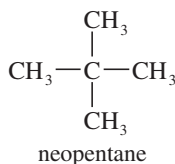
The O—H bond is more polarized than the N—H bond because O is more electronegative than N.

- The more hydrogen-bonding groups a molecule has, the higher its boiling point. The following examples all have approximately the same molecular weight and degree of branching.



SOLVED PROBLEM 2-2

Rank the following compounds in order of increasing boiling points. Explain the reasons for your chosen order.



SOLUTION

To predict relative boiling points, we should look for differences in (1) hydrogen bonding, (2) molecular weight and surface area, and (3) dipole moments. Except for neopentane, these compounds have similar molecular weights. The two alcohols (-ol suffix) engage in hydrogen bonding, so they should be the highest-boiling compounds. Unbranched pentan-1-ol should boil at a higher temperature than branched 2-methylbutan-2-ol, which has less surface area for van der Waals forces.

Among the other compounds, neopentane has the smallest molecular weight, and it is a compact spherical structure that minimizes van der Waals attractions. Neopentane should be the lowest-boiling compound.

Comparing the last two compounds, 2,3-dimethylbutane is more highly branched (and has a smaller surface area) than hexane. Therefore, 2,3-dimethylbutane should have a lower boiling point than hexane. We predict the following order:



10 °C

58 °C

69 °C

102 °C

138 °C

The actual boiling points are given here to show that our prediction is correct.

PROBLEM 2-6

For each pair of compounds, circle the compound you expect to have the higher boiling point. Explain your reasoning.

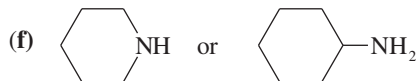
(a) $(\text{CH}_3)_3\text{C}-\text{C}(\text{CH}_3)_3$ or $(\text{CH}_3)_2\text{CH}-\text{CH}_2\text{CH}_2-\text{CH}(\text{CH}_3)_2$

(b) $\text{CH}_3(\text{CH}_2)_6\text{CH}_3$ or $\text{CH}_3(\text{CH}_2)_5\text{CH}_2\text{OH}$

(c) $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$ or $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$

(d) $\text{HOCH}_2-(\text{CH}_2)_4-\text{CH}_2\text{OH}$ or $(\text{CH}_3)_3\text{CCH}(\text{OH})\text{CH}_3$

(e) $(\text{CH}_3\text{CH}_2\text{CH}_2)_2\text{NH}$ or $(\text{CH}_3\text{CH}_2)_3\text{N}$

**PROBLEM-SOLVING HINT**

To predict relative boiling points, look for differences in

1. hydrogen bonding,
2. molecular weight and surface area, and
3. dipole moments.

2-3 Polarity Effects on Solubilities

In addition to affecting boiling points and melting points, intermolecular forces determine the solubility properties of organic compounds. The general rule is that “*like dissolves like*.” Polar substances dissolve in polar solvents, and nonpolar substances dissolve in nonpolar solvents. We discuss the reasons for this rule now, then apply the rule in later chapters when we discuss the solubility properties of organic compounds.

We consider four different cases: (1) a polar solute with a polar solvent, (2) a polar solute with a nonpolar solvent, (3) a nonpolar solute with a nonpolar solvent, and (4) a nonpolar solute with a polar solvent. We use sodium chloride and water as examples of polar solutes and solvents and paraffin “wax” and gasoline as examples of nonpolar solutes and solvents.

Polar Solute in a Polar Solvent (Dissolves) When you think about sodium chloride dissolving in water, it seems remarkable that the oppositely charged ions can be separated from each other. A great deal of energy is required to separate these ions. A polar solvent (such as water) can separate the ions because it *solvates* them (Figure 2-6). If water is the solvent, the solvation process is called *hydration*. As the salt dissolves, water molecules surround each ion, with the appropriate end of the water dipole moment next to the ion. The oxygen atoms of the water molecules approach the positively charged sodium ions. Water’s hydrogen atoms approach the negatively charged chloride ions.

Because water molecules are strongly polar, a large amount of energy is released when the sodium and chloride ions are hydrated. This energy is nearly sufficient to overcome the lattice energy of the crystal. The salt dissolves, partly because of strong

Application: Biochemistry

Most vitamins contain charged groups, making them water-soluble. As a result, they are rapidly eliminated and generally nontoxic. Vitamins A and D, however, are nonpolar and are stored in the fat tissue of the body, which is also nonpolar. Hence, these two vitamins are potentially toxic in large doses.

FIGURE 2-6 Polar solute in water (a polar solvent). The hydration of sodium and chloride ions by water molecules overcomes the lattice energy of sodium chloride. The salt dissolves.

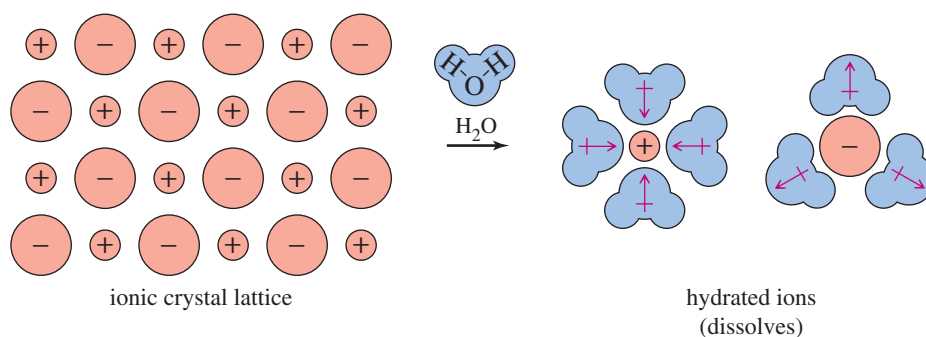
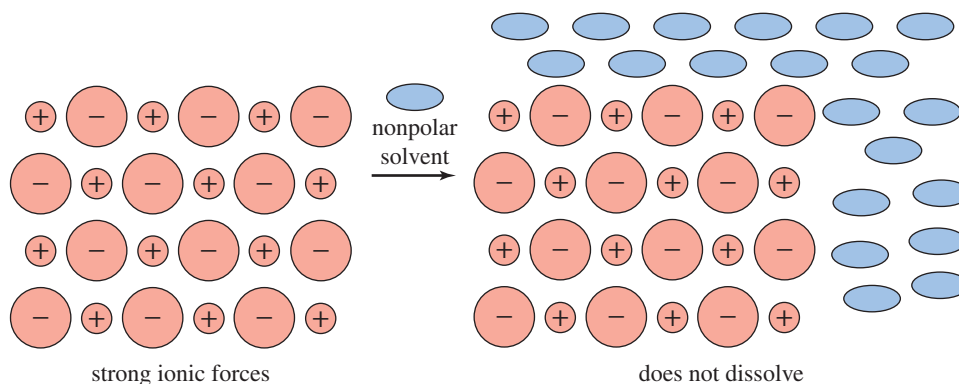
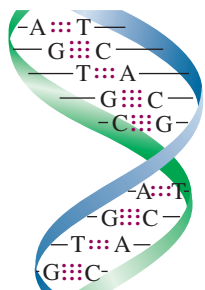


FIGURE 2-7 Polar solute in a nonpolar solvent. The intermolecular attractions of polar substances are stronger than their attractions for nonpolar solvent molecules. Thus, a polar substance does not dissolve in a nonpolar solvent.



Application: Biochemistry

Hydrogen bonding is essential for the structural integrity of many biological molecules. For example, the double helix structure of DNA is maintained, in part, by hydrogen bonds between the bases: Adenine (A) pairs with thymine (T), joined by two hydrogen bonds, and guanine (G) pairs with cytosine (C), joined by three hydrogen bonds. In the diagram below, the hydrogen bonds are represented by lines of red dots.



solvation by water molecules and partly because of the increase in entropy (randomness or freedom of movement) when it dissolves.

Polar Solute in a Nonpolar Solvent (Does Not Dissolve) If you stir sodium chloride with a nonpolar solvent such as turpentine or gasoline, you will find that the salt does not dissolve (Figure 2-7). The nonpolar molecules of these solvents do not solvate ions very strongly, and they cannot overcome the large lattice energy of the salt crystal. This is a case where the attractions of the ions in the solid for each other are much greater than their attractions for the solvent.

Nonpolar Solute in a Nonpolar Solvent (Dissolves) Paraffin “wax” dissolves in gasoline. Both paraffin and gasoline are mixtures of nonpolar hydrocarbons (Figure 2-8). The molecules of a nonpolar substance (paraffin) are weakly attracted to each other, and these van der Waals attractions are easily overcome by van der Waals attractions with the solvent. Although there is little change in energy when the nonpolar substance dissolves in a nonpolar solvent, there is a large increase in entropy.

Nonpolar Solute in a Polar Solvent (Does Not Dissolve) Nonpolar solids such as paraffin “wax” and other waxes do not dissolve in a polar solvent such as water. Why not? The nonpolar molecules are only weakly attracted to each other, and little energy is required to separate them. The problem is that the water molecules are strongly attracted to each other by their hydrogen bonding.

If a nonpolar paraffin molecule were to dissolve, the water molecules around it would have to form a cavity. Water molecules at the edge of the cavity have fewer available neighbors for hydrogen bonding, resulting in a tighter, more rigid, ice-like structure around the cavity. This tighter structure results in an unfavorable decrease in entropy (Figure 2-9). Nonpolar substances that do not dissolve in water are called **hydrophobic** (“water-hating”).

Figures 2-6 through 2-9 show why the saying “like dissolves like” is generally true. Polar substances dissolve in polar solvents, and nonpolar substances dissolve in nonpolar solvents. This general rule also applies to the mixing of liquids. For example,

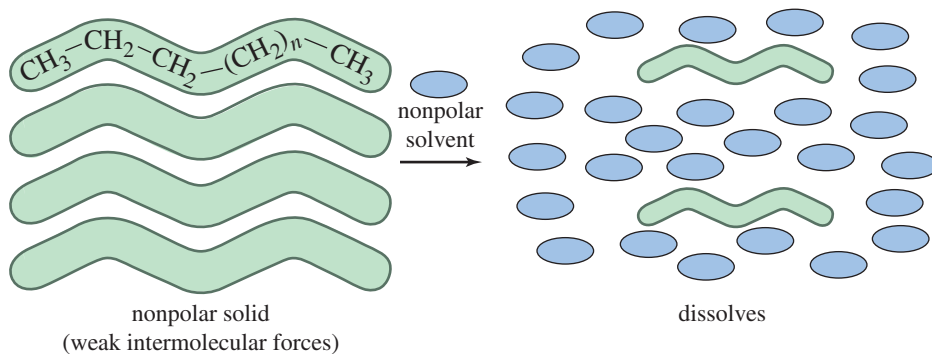


FIGURE 2-8 Nonpolar solute in a nonpolar solvent. The weak intermolecular attractions of a nonpolar substance are overcome by the weak attractions for a nonpolar solvent. The nonpolar substance dissolves.

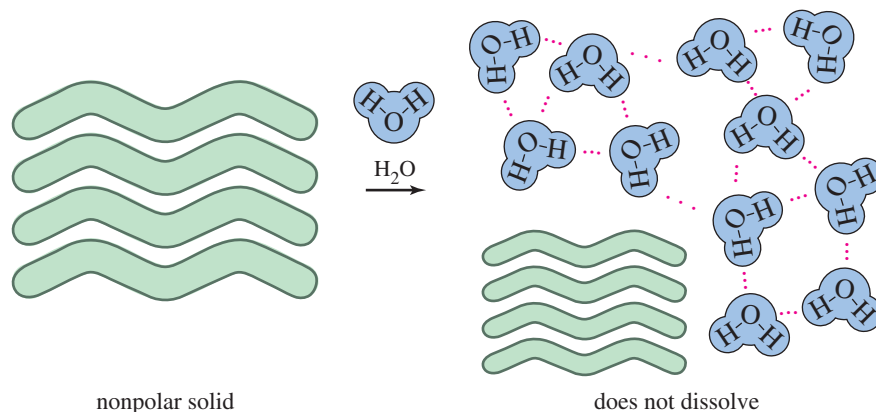


FIGURE 2-9 Nonpolar solute in a polar solvent (water). Nonpolar substances are hydrophobic. They do not dissolve in water because of the unfavorable entropy effects associated with forming a hydrogen-bonded shell of water molecules around a nonpolar molecule.

water and gasoline (or oil) do not mix. Gasoline and oil are both hydrophobic, nonpolar hydrocarbons, however, and they mix freely with each other.

Ethanol is a polar molecule, and it is miscible with water; that is, it mixes freely with water in all proportions. Polar substances that readily dissolve in water are called **hydrophilic** (“water-loving”). Ethanol has an O—H group that forms hydrogen bonds with water molecules. When ethanol dissolves in water (Figure 2-10), it forms new ethanol–water hydrogen bonds to replace the water–water and ethanol–ethanol hydrogen bonds that are broken:

In Sections 2-15 through 2-17, we will see many kinds of organic compounds with a wide variety of functional groups. As you encounter these new compounds, you should look to see whether the molecules are polar or nonpolar and whether they can engage in hydrogen bonding. Those that are nonpolar tend to be hydrophobic. Those that are strongly polar may be hydrophilic, especially if they can engage in hydrogen bonding.



Motor oil and water do not mix because the nonpolar oil molecules cannot displace the strong intermolecular attractions between water molecules.

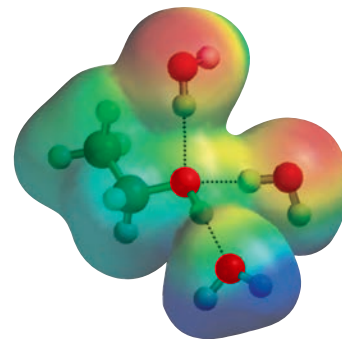
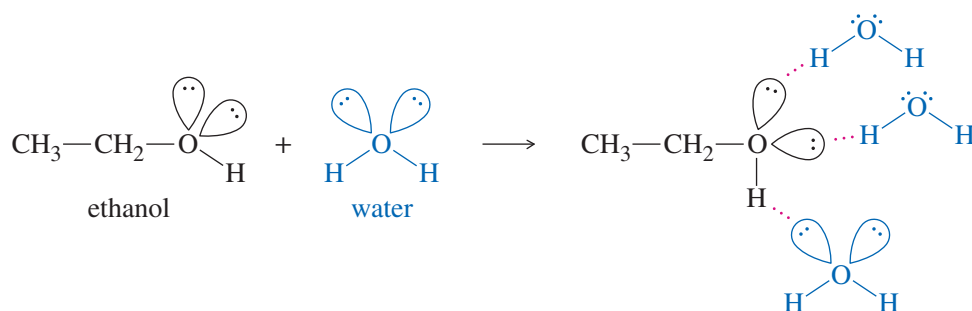


FIGURE 2-10 Ethanol is hydrophilic because its OH groups form hydrogen bonds with water.

FOCUS Hydrogen Bonding and Solubility in Water

Solubility is a measure of how much of one substance dissolves in another. Solubility in water is a critical property in chemistry labs, environmental considerations, and medical applications.

Hydrocarbons with no polar atoms have very small solubility in water. For example, hexane has a solubility of 0.001 g/100 mL H_2O . These compounds are called *insoluble*, although that does not mean the same as zero solubility.

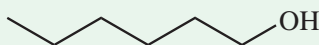
The most important factor determining a compound's solubility in water is the presence of atoms (O and N) that can form hydrogen bonds with water, and the ratio of these atoms to the number of carbons in the molecule. As a rough approximation, one O or N forms hydrogen bonds strong enough to allow about four carbons to dissolve in water. These trends are shown in the compounds below.

- Increasing the number of hydrogen-bonding groups with the same number of C increases solubility.



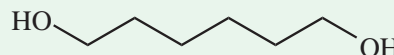
hexane

sol. = 0.001 g/100 mL H_2O
or, sol. = insoluble



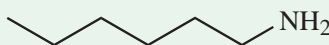
hexan-1-ol

sol. = 0.59 g/100 mL H_2O



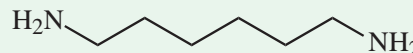
hexane-1,6-diol

sol. = 50 g/100 mL H_2O



hexan-1-amine

sol. = 1.2 g/100 mL H_2O



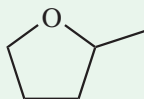
hexane-1,6-diamine

sol. = 49 g/100 mL H_2O

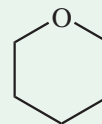
- Increasing the number of carbons while keeping the number of hydrogen-bonding groups constant decreases solubility.

tetrahydrofuran, $\text{C}_4\text{H}_8\text{O}$

sol. = miscible in H_2O

2-methyltetrahydrofuran, $\text{C}_5\text{H}_{10}\text{O}$

sol. = 15 g/100 mL H_2O

tetrahydropyran, $\text{C}_5\text{H}_{10}\text{O}$

sol. = 8 g/100 mL H_2O

Although halogen atoms are electronegative, they do not participate in hydrogen bonding in organic compounds and do not increase water solubility.

PROBLEM-SOLVING HINT

Hydrogen bonding is the most important factor in whether a compound is hydrophilic. In general, one hydrogen-bonding polar group can carry about four carbons into water. Most hydrogen-bonding compounds with three or fewer carbons are miscible with water. Multiple hydrogen-bonding groups in a molecule increase its solubility in water.

PROBLEM 2-7

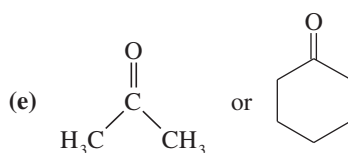
Circle the member of each pair that is more soluble in water.

(a) $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$ or $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$

(b) $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$ or $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$

(c) $\text{CH}_3\text{CH}_2\text{NHCH}_3$ or $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$

(d) $\text{CH}_3\text{CH}_2\text{OH}$ or $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$

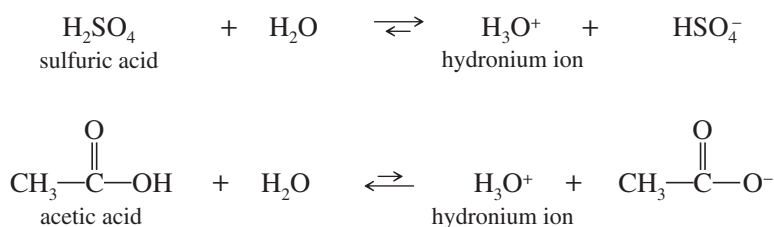


2-4 Arrhenius Acids and Bases

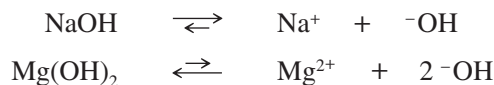
The properties and reactions of acids and bases are central to our study of organic chemistry. We need to consider exactly what is meant by the terms **acid** and **base**. Most people would agree that H_2SO_4 is an acid and NaOH is a base. Is BF_3 an acid or a base? Is ethylene ($\text{H}_2\text{C}=\text{CH}_2$) an acid or a base? To answer these questions, we need to understand the three different definitions of acids and bases: the Arrhenius definition, the Brønsted–Lowry definition, and the Lewis definition.

Acidic compounds were first classified on the basis of their sour taste. The Latin terms *acidus* (sour) and *acetum* (vinegar) gave rise to our modern terms *acid* and *acetic acid*. Alkaline compounds (bases) were substances that neutralize acids, such as limestone and plant ashes (*al kalai* in Arabic).

In 1887, Swedish chemist Svante Arrhenius introduced the **Arrhenius** theory, which defines acids as *substances that dissociate in water to give hydronium ions*. The stronger acids, such as sulfuric acid (H_2SO_4), were assumed to dissociate to a greater degree than weaker acids, such as acetic acid (CH_3COOH).



According to the Arrhenius definition, bases are *substances that dissociate in water to give hydroxide ions*. Strong bases, such as NaOH , were assumed to dissociate more completely than weaker, sparingly soluble bases such as $\text{Mg}(\text{OH})_2$.



The acidity or basicity of an aqueous (water) solution is measured by the concentration of H_3O^+ . This value also implies the concentration of $\text{}^-\text{OH}$ because these two concentrations are related by the water ion-product constant:

$$K_w = [\text{H}_3\text{O}^+][\text{}^-\text{OH}] = 1.00 \times 10^{-14} \text{ } M^2 \text{ (at } 25^\circ\text{C)}$$

In a neutral solution, the concentrations of H_3O^+ and $\text{}^-\text{OH}$ are equal.

$$[\text{H}_3\text{O}^+] = [\text{}^-\text{OH}] = 1.00 \times 10^{-7} \text{ } M \text{ in a neutral solution}$$

Acidic and basic solutions are defined by an excess of H_3O^+ or $\text{}^-\text{OH}$.

$$\text{acidic: } [\text{H}_3\text{O}^+] > 10^{-7} \text{ } M \text{ and } [\text{}^-\text{OH}] < 10^{-7} \text{ } M$$

$$\text{basic: } [\text{H}_3\text{O}^+] < 10^{-7} \text{ } M \text{ and } [\text{}^-\text{OH}] > 10^{-7} \text{ } M$$

Because these concentrations can span a wide range of values, the acidity or basicity of a solution is usually measured on a logarithmic scale. The **pH** is defined as the negative logarithm (base 10) of the H_3O^+ concentration.

$$\text{pH} = -\log_{10}[\text{H}_3\text{O}^+]$$

A neutral solution has a pH of 7, an acidic solution has a pH less than 7, and a basic solution has a pH greater than 7.

Svante August Arrhenius

(1859–1927) was a Swedish physical chemist who pioneered the field of chemical kinetics. He was a pioneer in research on atmospheric carbon dioxide in relation to the rising surface temperature on Earth, which is now known as the Greenhouse Effect. In recognizing the extraordinary service, he rendered to the advancement of chemistry by his electrolytic theory of dissociation, Arrhenius was awarded the Nobel Prize in Chemistry (1903).

PROBLEM 2-8

Calculate the pH of the following solutions.

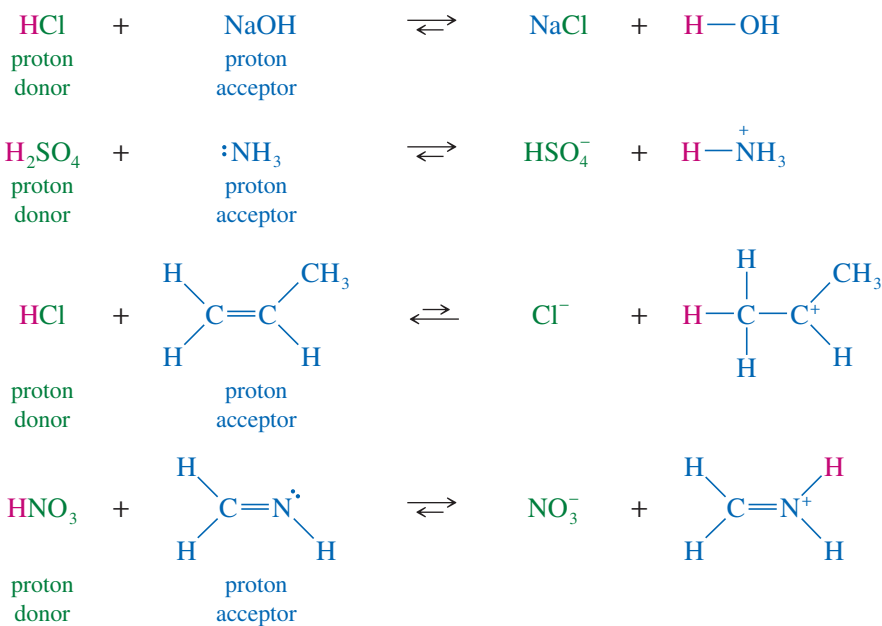
- (a) 5.00 g of HBr in 100 mL of aqueous solution
 (b) 1.50 g of NaOH in 50 mL of aqueous solution

The Arrhenius definition was an important contribution to understanding many acids and bases, but it does not explain why a compound such as ammonia (NH_3) neutralizes acids, even though it has no hydroxide ion in its molecular formula. In the next section, we discuss a more versatile theory of acids and bases that includes ammonia and a wider variety of organic acids and bases.

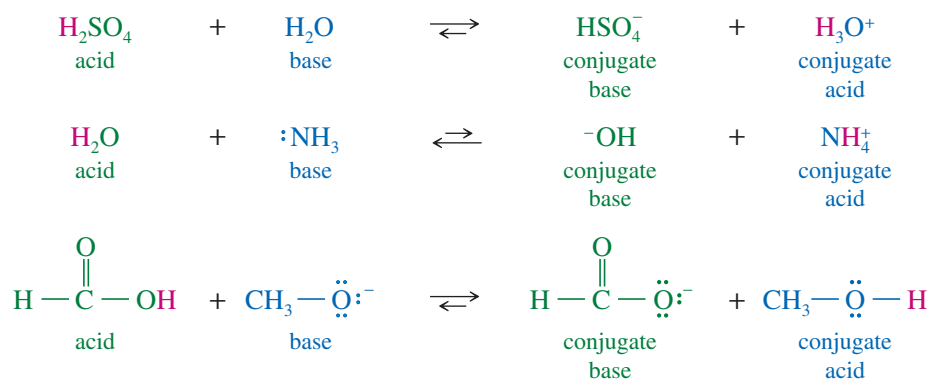
2-5 Brønsted–Lowry Acids and Bases

In 1923, Danish chemist Johannes Brønsted and English chemist Thomas Lowry independently proposed defining acids and bases by following the transfer of protons. A **Brønsted–Lowry acid** is any species that can donate a proton, and a **Brønsted–Lowry base** is any species that can accept a proton. These definitions also include all the Arrhenius acids and bases because compounds that dissociate to give H_3O^+ are proton donors, and compounds that dissociate to give OH^- are proton acceptors. (Hydroxide ion accepts a proton to form H_2O .)

In addition to Arrhenius acids and bases, the Brønsted–Lowry definition includes bases that have no hydroxide ions, yet can accept protons. Consider the following examples of acids donating protons to bases. NaOH is a base under either the Arrhenius or Brønsted–Lowry definition. The other three are Brønsted–Lowry bases but not Arrhenius bases, because they have no hydroxide ions.



When a base accepts a proton, it becomes an acid capable of returning that proton. When an acid donates its proton, it becomes a base capable of accepting that proton back. One of the most important principles of the Brønsted–Lowry definition is this concept of **conjugate acids and bases**. For example, NH_4^+ and NH_3 are a conjugate acid–base pair. NH_3 is the base; when it accepts a proton, it is transformed into its conjugate acid, NH_4^+ . Many compounds (water, for instance) can react as either an acid or a base. Here are some additional examples of conjugate acid–base pairs.

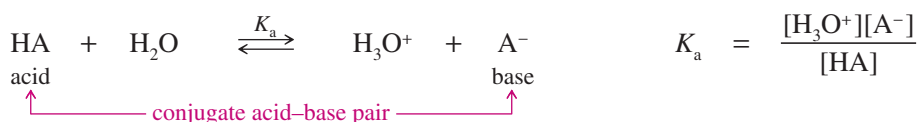


2-6 Strengths of Acids and Bases

It seems obvious that the strength of an acid should reflect its tendency to protonate a base by donating a proton. Similarly, the strength of a base should reflect its tendency to deprotonate an acid by accepting a proton. But we need quantitative values to compare different acids and bases. For example, if we add both sulfuric acid and acetic acid to water, which one will protonate water? If we add water to ammonium sulfate, will water act as an acid or a base? The theory of acid strength and base strength allows us to answer these questions with confidence.

2-6A Acid Strength

The strength of a Brønsted–Lowry acid is expressed as it is in the Arrhenius definition, by the extent of its ionization in water. The general reaction of an acid (HA) with water is as follows:



K_a is called the **acid-dissociation constant**, and its value indicates the relative strength of the acid. The stronger the acid, the more it dissociates, giving a larger value of K_a . Acid-dissociation constants vary over a wide range. Strong acids are almost completely ionized in water, and their dissociation constants are greater than 1. Most organic acids are weak acids, with K_a values less than 10^{-4} . Many organic compounds are extremely weak acids; for example, methane and ethane are essentially with K_a values less than 10^{-40} .

Because they span such a wide range, acid-dissociation constants are often expressed on a logarithmic scale. The $\text{p}K_a$ of an acid is defined just like the pH of a solution: as the negative logarithm (base 10) of K_a .

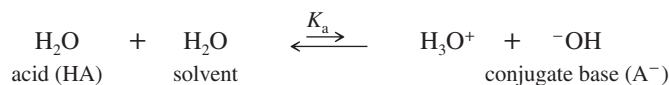
$$\text{p}K_a = -\log_{10} K_a$$

SOLVED PROBLEM 2-3

Calculate K_a and $\text{p}K_a$ for water.

SOLUTION

The equilibrium that defines K_a for water is



(continued)

PROBLEM-SOLVING HINT

Students sometimes forget that the K_a of water ($1.8 \times 10^{-16} M$) is different from the ion-product constant $K_w = 1.00 \times 10^{-14} M^2$. In calculating the K_a , you must divide by the concentration of the acid (water), which is 55.6 moles per liter in pure water.

Water serves as both the acid and the solvent in this dissociation. The equilibrium expression is

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]} = \frac{[\text{H}_3\text{O}^+][\text{OH}^-]}{[\text{H}_2\text{O}]}$$

We already know that $[\text{H}_3\text{O}^+][\text{OH}^-] = K_w = 1.00 \times 10^{-14} \text{ M}^2$, the ion-product constant for water. The concentration of H_2O in water is simply the number of moles of water in 1 L (about 1 kg).

$$\frac{1000 \text{ g/L}}{18.0 \text{ g/mol}} = 55.6 \text{ mol/L}$$

Substitution gives

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{OH}^-]}{[\text{H}_2\text{O}]} = \frac{1.00 \times 10^{-14}}{55.6} = 1.8 \times 10^{-16} \text{ M}$$

The logarithm of 1.8×10^{-16} is -15.7 , so the $\text{p}K_a$ of water is 15.7 .

Strong acids generally have $\text{p}K_a$ values around 0 (or even negative), and weak acids, such as most organic acids, have $\text{p}K_a$ values that are greater than 4. *Weaker acids have larger $\text{p}K_a$ values.* Table 2-2 lists K_a and $\text{p}K_a$ values for some common inorganic and organic compounds. Notice that the $\text{p}K_a$ values increase as the K_a values decrease. A much larger table of $\text{p}K_a$ values appears in Appendix 4.

PROBLEM-SOLVING HINT

In most cases, the $\text{p}K_a$ of an acid corresponds to the pH where the acid is about half dissociated. At a lower (more acidic) pH, the acid is mostly undissociated; at a higher (more basic) pH, the acid is mostly dissociated.

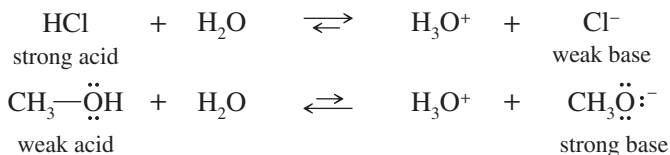
PROBLEM 2-9

Ammonia appears in Table 2-2 as both an acid and a conjugate base.

- Explain how ammonia can act as both an acid and a base. Which of these roles does it commonly fill in aqueous solutions?
- Show how water can serve as both an acid and a base.
- Calculate K_a and $\text{p}K_a$ for the hydronium ion, H_3O^+ .
- Show how methanol (CH_3OH) can serve as both an acid and a base. Write an equation for the reaction of methanol with sulfuric acid.

2-6B Base Strength

The strength of an acid is inversely related to the strength of its conjugate base. For an acid (HA) to be strong, its conjugate base (A^-) must be stable in its anionic form; otherwise, HA would not easily lose its proton. Therefore, the conjugate base of a strong acid must be a weak base. On the other hand, when an acid is weak, its conjugate base is a strong base.



In the reaction of an acid with a base, the equilibrium generally favors the *weaker* acid and base. For example, in the preceding reactions, H_3O^+ is a weaker acid than HCl but a stronger acid than CH_3OH . It also follows that H_2O is a stronger base than Cl^- but a weaker base than CH_3O^- .

The strength of a base is measured much like the strength of an acid, by using the equilibrium constant of the hydrolysis reaction.

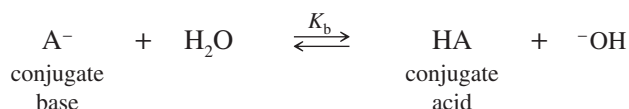


TABLE 2-2 Relative Strength of Some Common Organic and Inorganic Acids and Their Conjugate Bases

	Acid		Conjugate Base	K_a	pK_a
strong acids	HCl hydrochloric acid	$+ \text{H}_2\text{O} \rightleftharpoons$	$\text{H}_3\text{O}^+ + \text{Cl}^-$ chloride ion	1×10^7	-7
	H_3O^+ hydronium ion	$+ \text{H}_2\text{O} \rightleftharpoons$	$\text{H}_3\text{O}^+ + \text{H}_2\text{O}$ water	55.6	-1.7
	HF hydrofluoric acid	$+ \text{H}_2\text{O} \rightleftharpoons$	$\text{H}_3\text{O}^+ + \text{F}^-$ fluoride ion	6.8×10^{-4}	3.17
weak acids	$\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ formic acid	$+ \text{H}_2\text{O} \rightleftharpoons$	$\text{H}_3\text{O}^+ + \text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}^-$ formate ion	1.7×10^{-4}	3.76
	$\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ acetic acid	$+ \text{H}_2\text{O} \rightleftharpoons$	$\text{H}_3\text{O}^+ + \text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}^-$ acetate ion	1.8×10^{-5}	4.74
	$\text{H}-\text{C}\equiv\text{N}:$ hydrocyanic acid	$+ \text{H}_2\text{O} \rightleftharpoons$	$\text{H}_3\text{O}^+ + :\text{C}\equiv\text{N}^-$ cyanide ion	6.0×10^{-10}	9.22
	$^+\text{NH}_4$ ammonium ion	$+ \text{H}_2\text{O} \rightleftharpoons$	$\text{H}_3\text{O}^+ + :\text{NH}_3$ ammonia	5.8×10^{-10}	9.24
	H_2O water	$+ \text{H}_2\text{O} \rightleftharpoons$	$\text{H}_3\text{O}^+ + \text{HO}^-$ hydroxide ion	1.8×10^{-16}	15.7
very weak	$\text{CH}_3\text{CH}_2-\text{OH}$ ethyl alcohol	$+ \text{H}_2\text{O} \rightleftharpoons$	$\text{H}_3\text{O}^+ + \text{CH}_3\text{CH}_2\text{O}^-$ ethoxide ion	1.3×10^{-16}	15.9
	NH_3 ammonia	$+ \text{H}_2\text{O} \rightleftharpoons$	$\text{H}_3\text{O}^+ + :\ddot{\text{N}}\text{H}_2^-$ amide ion	10^{-36}	36
not acidic	CH_4 methane	$+ \text{H}_2\text{O} \rightleftharpoons$	$\text{H}_3\text{O}^+ + :\text{CH}_3^-$ methyl anion	$<10^{-50}$	50

The equilibrium constant (K_b) for this reaction is called the *base-dissociation constant* for the base A^- . Because this constant spans a wide range of values, it is often given in logarithmic form. The negative logarithm (base 10) of K_b is defined as pK_b .

$$K_b = \frac{[\text{HA}][\text{OH}^-]}{[\text{A}^-]} \quad pK_b = -\log_{10} K_b$$

When we multiply K_a by K_b , we can see how the acidity of an acid is related to the basicity of its conjugate base.

$$(K_a)(K_b) = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]} \frac{[\text{HA}][\text{OH}^-]}{[\text{A}^-]} = [\text{H}_3\text{O}^+][\text{OH}^-] = K_w = 1.0 \times 10^{-14}$$

water ion-product constant

$$(K_a)(K_b) = 10^{-14}$$

Application: Drugs

The acid-base properties of many natural products are important for their isolation, distribution in the body, and therapeutic effects. For example, morphine (page 38) is isolated from the opium poppy and crosses into the brain as the free base, where the nitrogen is not charged. However, it exerts its pain-relieving effects as the protonated, charged species.

Logarithmically,

$$\text{p}K_{\text{a}} + \text{p}K_{\text{b}} = -\log 10^{-14} = 14$$

In an aqueous solution, the product of K_{a} and K_{b} must always equal the ion-product constant of water, $K_{\text{w}} = 10^{-14}$. If the value of K_{a} is large, the value of K_{b} must be small; that is, the stronger an acid, the weaker its conjugate base. Similarly, a small value of K_{a} (weak acid) implies a large value of K_{b} (strong base).

A stronger acid has a weaker conjugate base.
 A weaker acid has a stronger conjugate base.
 Acid–base reactions favor the weaker acid and the weaker base.

2-7 Equilibrium Positions of Acid–Base Reactions

A Brønsted–Lowry acid–base reaction transfers a positively charged proton from an acid to a base. The products are the conjugate base of the original acid and the conjugate acid of the original base. For the simple case where the acids are uncharged and the bases have negative charges, this is the general equation:



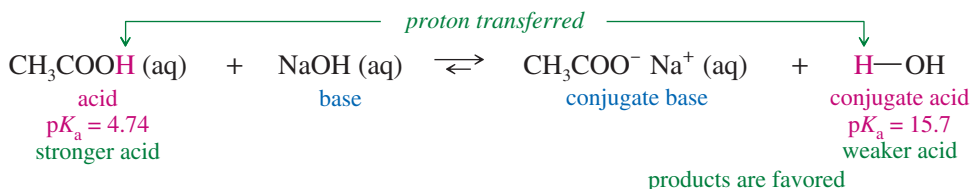
How do we know whether the reaction proceeds from left to right as written? Another way of asking that question is “Does the reaction favor the products at equilibrium?” How do we know which way the reaction goes and which species are favored by the equilibrium?

We can show the following three principles mathematically:

1. The acid–base equilibrium favors formation of the weaker acid and the weaker base.
2. The weaker acid has the larger $\text{p}K_{\text{a}}$. The weaker base has the larger $\text{p}K_{\text{b}}$.
3. The weaker acid and the weaker base are always on the same side of the equation. Both of them are reactants, or both of them are products.

We can predict which side of the equation is favored by comparing the strength of the two acids or by comparing the strength of the two bases. Either comparison gives the same result. The weaker acid is the one with the smaller K_{a} , which implies a larger $\text{p}K_{\text{a}}$. The weaker base is the one with the smaller K_{b} , which implies a larger $\text{p}K_{\text{b}}$. If we know the $\text{p}K_{\text{a}}$ values or the $\text{p}K_{\text{b}}$ values, then we can easily predict which side of the equation is favored.

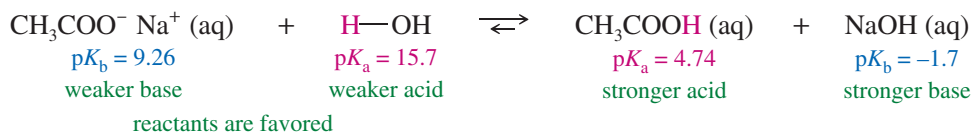
For example, consider the reaction of acetic acid with sodium hydroxide in water. We need to identify which reactant is the acid, which is the base, and which proton is transferred. Once those are identified, we can predict the products.



From Table 2-2 we know that the $\text{p}K_{\text{a}}$ for acetic acid is 4.74, and that the $\text{p}K_{\text{a}}$ for water is 15.7. Water is the weaker acid by about 11 $\text{p}K_{\text{a}}$ units. The equilibrium for this

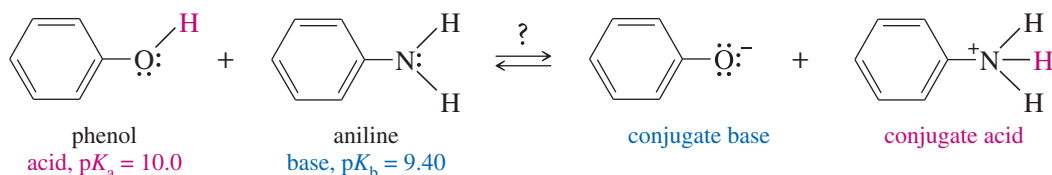
reaction lies far to the right and favors the products by 11 pK_a units. We can confidently say that this reaction goes to completion.

We could write this equation backward, interchanging the products and reactants. In that case, we would say that the reaction does not proceed, or that the equilibrium lies far to the left and favors the reactants.

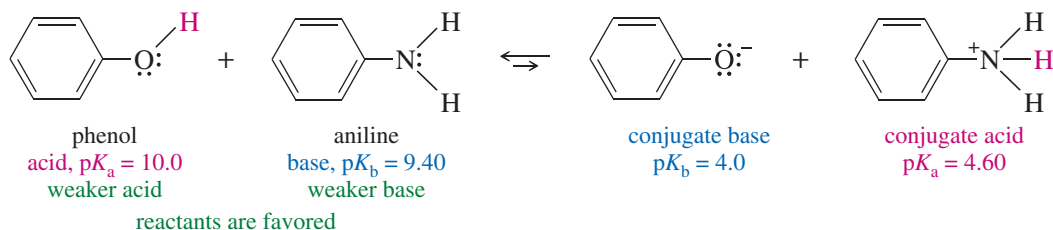


Note that in this equation, the pK_b values of the bases have been included simply by subtracting the pK_a values from 14 (K_w , the ion-product constant for water): $pK_b = 14 - pK_a$ of the conjugate acid. If we had compared the pK_b values instead of the pK_a values, we would have obtained the same result.

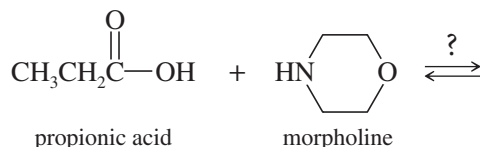
Consider another example: the possible acid–base reaction between phenol and aniline. Phenol is a weak acid with $pK_a = 10.0$, and aniline, with $pK_b = 9.4$, is a weak base. Does this reaction go toward the left or the right?



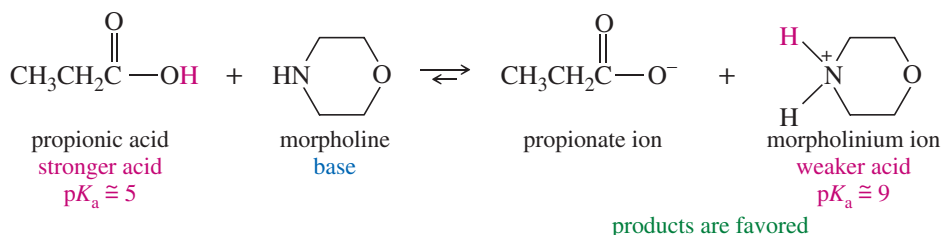
To compare pK_a or pK_b values, we must first subtract from 14 to get the pK_a and pK_b of the products. We find that the conjugate base of phenol has $pK_b = 4.0$, and that the conjugate acid of aniline has $pK_a = 4.60$. The equilibrium lies far to the left, and this reaction will not go as written. The reverse reaction would go spontaneously.



Predicting the position of an acid–base equilibrium becomes more challenging when no values are available for pK_a or pK_b . In most cases, we can estimate approximate values of pK_a or pK_b by comparing the compounds with similar compounds listed in the tables. For example, we can predict the following equilibrium using only the values in Table 2-2:



Propionic acid is a carboxylic acid, like acetic acid. The pK_a of acetic acid is 4.74, so we estimate the pK_a of propionic acid to be about 5. Morpholine should be a weak base like ammonia. We predict that propionic acid might lose a proton to morpholine. Table 2-2 shows that the pK_a of protonated ammonia (the ammonium ion, NH_4^+) is 9.26, so we estimate the pK_a of protonated morpholine to be about 9.



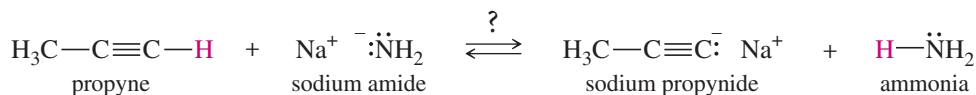
The products contain the weaker acid, so the products are favored. We can be fairly confident of this result even though we used estimated pK_a values, because there is a large difference in pK_a between the reactants and the products. (The literature values of pK_a are 4.87 for propionic acid and 8.4 for the morpholinium ion.)

PROBLEM-SOLVING STRATEGY Predicting Acid–Base Equilibrium Positions

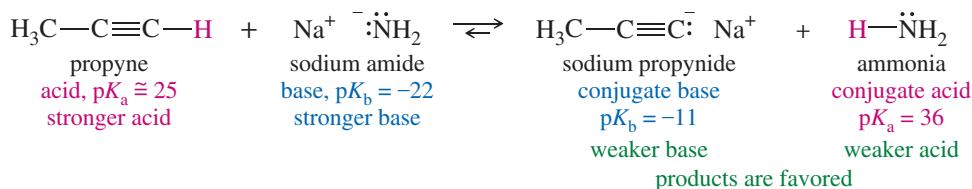
The preceding examples show how we can confidently predict the favored products or reactants in most acid–base reactions. Let's summarize the steps involved and then use these steps to work through a particularly difficult problem. The steps we used above (sometimes combined) are these:

- Step 1:** Complete the reaction (if necessary), and label the acids and bases on each side.
Step 2: Assign pK_a or pK_b values from tables or by analogy with similar compounds. Note that Appendix 4 gives an extensive list of pK_a values.
Step 3: Calculate any needed pK_a or pK_b values using the fact that $pK_a + pK_b = 14$ for a conjugate acid–base pair.
Step 4: Choose the side with the weaker acid or weaker base (larger value of pK_a or pK_b) as favored. The difference in pK_a or pK_b values shows how strongly that side is favored.

As an example, the following reaction is a strongly basic one that cannot take place in water because water would protonate the strong bases involved. In fact, this reaction is usually done in liquid ammonia solution.



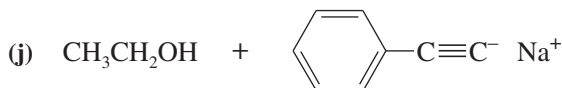
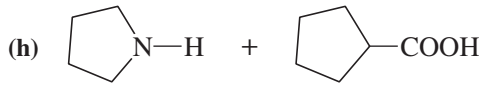
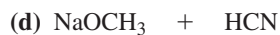
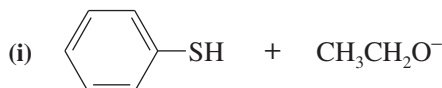
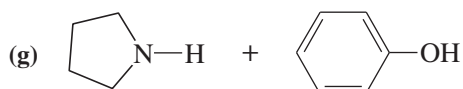
- Step 1:** In this reaction, a very weakly acidic proton is transferred from propyne to the amide ion (NH_2^-) to give the propynide anion and ammonia. Propyne serves as a weak acid, and ammonia is the conjugate acid of the amide ion, a powerful base.
Step 2: The pK_a of ammonia is shown in Table 2-2 and Appendix 4 as about 36. When a pK_a is this large, it depends on the solvent, and it is difficult to measure. The pK_a of propyne is not listed in Table 2-2 or in Appendix 4, but we can estimate it. Appendix 4 lists the pK_a of acetylene, $\text{H}-\text{C}\equiv\text{C}-\text{H}$, as 25. This should be a good approximation for the pK_a of propyne.
Step 3: The pK_b of the amide anion can be found by subtracting the pK_a of ammonia from 14: $14 - 36 = -22$. This large, negative pK_b indicates an unusually powerful base. The pK_b of the propynide ion is $14 - 25 = -11$. This is also a very strong base, but not as strong as the amide ion.



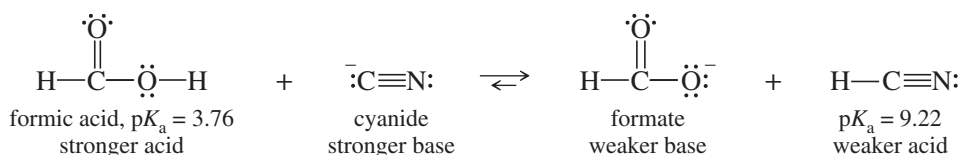
- Step 4:** We can compare the pK_a values or the pK_b values. Either way, the products are the weaker acid and the weaker base, by about 11 pK_a units. The products are strongly favored in this reaction, and the reaction should proceed as written.

PROBLEM 2-10 (PARTIALLY SOLVED)

Write equations for the following acid–base reactions. Use the information in Table 2-2 or Appendix 4 to predict whether the equilibrium will favor the reactants or the products.

**SOLUTION**

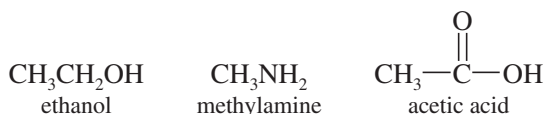
to (a): Cyanide is the conjugate base of HCN. It can accept a proton from formic acid:



Reading from Table 2-2, formic acid ($pK_a = 3.76$) is a stronger acid than HCN ($pK_a = 9.22$), and cyanide is a stronger base than formate. The products (weaker acid and base) are favored.

PROBLEM 2-11

Ethanol, methylamine, and acetic acid are all amphoteric, reacting as either acids or bases depending on the conditions.



- (a) Rank ethanol, methylamine, and acetic acid in decreasing order of acidity. In each case, show the equation for the reaction with a generic base (B:^-) to give the conjugate base.
- (b) Rank ethanol, methylamine, and acetic acid in decreasing order of basicity. In each case, show the equation for the reaction with a generic acid (HA) to give the conjugate acid.

PROBLEM-SOLVING HINT

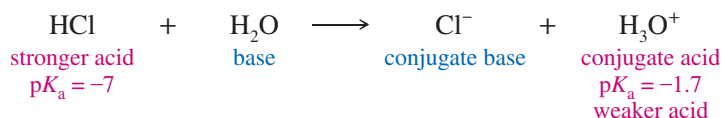
An acid will donate a proton to the conjugate base of any weaker acid (smaller K_a or higher pK_a).

PROBLEM-SOLVING HINT

Use either acid strengths or base strengths to determine the favored side of an acid–base equilibrium. The side with weaker acid (larger pK_a) or the weaker base (larger pK_b) is favored.

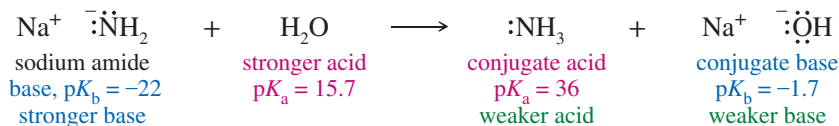
2-8 Solvent Effects on Acidity and Basicity

Water is amphoteric; it can react as both an acid and a base. Its conjugate acid is H_3O^+ , and its conjugate base is ^-OH . When water is the solvent for a reaction, it is present in large excess. Any acid that is stronger than H_3O^+ ($pK_a = -1.7$) does not survive in water, but reacts with water to give H_3O^+ . For example, if we add HCl gas ($pK_a = -7$) to water, it immediately reacts exothermically to give H_3O^+ , a weaker acid.



When we say that we add HCl (aq) to a reaction, we are actually adding H_3O^+ (aq) and Cl^- (aq) because HCl is too strong an acid to coexist with water. We should realize that we are not using an acid with $\text{p}K_a = -7$, but actually one with $\text{p}K_a = -1.7$.

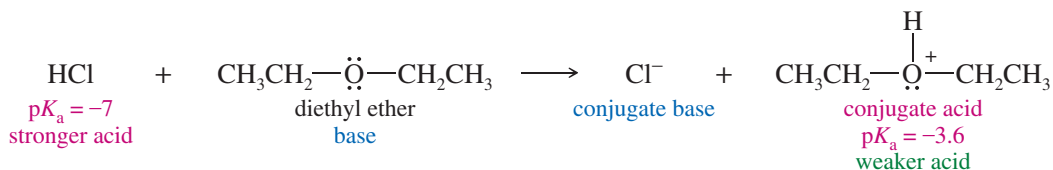
The same principle applies to bases that are stronger than OH^- , the conjugate base of water, with $\text{p}K_b = -1.7$. If we add some sodium amide (NaNH_2 , $\text{p}K_b = -22$) to water, it immediately reacts exothermically to give OH^- , a weaker base.



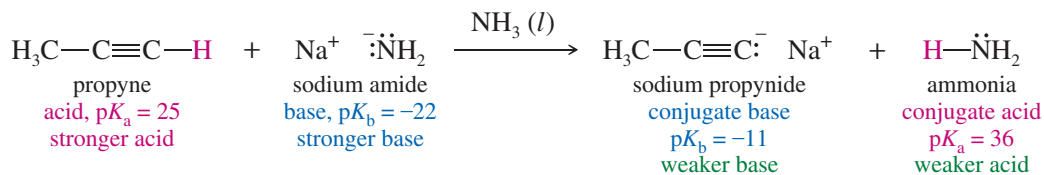
Thus, water narrows (“levels”) the possible ranges of acid strength and base strength to values of $\text{p}K_a > -1.7$ and values of $\text{p}K_b > -1.7$. Any acid with $\text{p}K_a$ more negative than -1.7 will be leveled to H_3O^+ , with $\text{p}K_a = -1.7$. Any base with $\text{p}K_b$ more negative than -1.7 will be leveled to OH^- , with $\text{p}K_b = -1.7$.

Alcohols ($\text{R}-\text{OH}$) react with both strong acids and strong bases just like water does. For example, strong acids protonate ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) to give protonated ethanol ($\text{CH}_3\text{CH}_2\text{OH}_2^+$), an acid with $\text{p}K_a = -2.4$. Any acid with a $\text{p}K_a$ that is more negative than -2.4 will donate a proton to ethanol. Strong bases deprotonate ethanol to give ethoxide ion ($\text{CH}_3\text{CH}_2\text{O}^-$), a base with $\text{p}K_b = -1.9$. Any base with a $\text{p}K_b$ that is more negative than -1.9 will become protonated in ethanol solution.

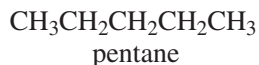
Other solvents may or may not narrow the range of acidity and basicity as much as water and alcohols do. For example, simple ethers tend to level acids but not bases. Diethyl ether ($\text{CH}_3\text{CH}_2-\text{O}-\text{CH}_2\text{CH}_3$) is not an acid, so it can be used as a solvent for some exceedingly strong bases. Diethyl ether is a weak base, however, so it narrows the range of acids. It reacts with any acid that is stronger than protonated diethyl ether ($\text{p}K_a = -3.6$).



Most ethers are not acidic, so they are often used as solvents for reagents that are stronger bases than hydroxide ions. Some basic reactions are best run in liquid ammonia, which has a $\text{p}K_a$ of 36, and accommodates bases with $\text{p}K_b > -22$. For example, reactions using sodium amide (the conjugate base of ammonia) are usually done in liquid ammonia, NH_3 (l).



Some organic solvents do not react appreciably with either acids or bases. Alkanes (saturated hydrocarbons) such as pentane and hexane are not acidic or basic, and we can use strong acids and bases in these solvents with little or no leveling of acidity or basicity.



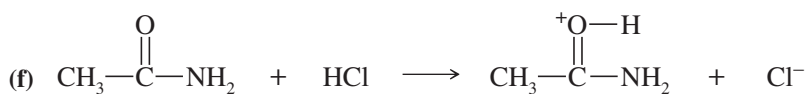
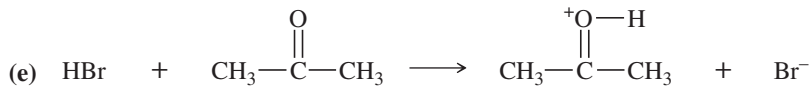
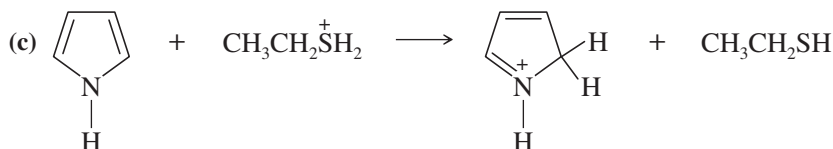
SUMMARY Acidity and Basicity Limitations in Common Solvents

Solvent	Structure	Typical pK_a Limit	Limiting Species	Typical pK_b Limit	Limiting Species
water	$H-O-H$	-1.7	H_3O^+	-1.7	HO^-
alcohols	$R-O-H$	-2.4	ROH_2^+	-1.9	RO^-
ethers	$R-O-R$	-3.6	$R-O^+(H)-R$	none	none
ammonia	NH_3	9.2	NH_4^+	-22	NH_2^-
alkanes	$R-H$	none	none	none	none

Values of pK_a and pK_b are defined using equilibrium reactions of the acids and bases in water. Values of pK_a between -1.7 (H_3O^+) and 15.7 (H_2O) and values of pK_b between -1.7 (OH^-) and 15.7 (H_2O) are measured in water, and are most accurate. Values outside this range cannot be measured in water because of water's leveling effect. There is no standard method for measuring these difficult values of pK_a and pK_b , and they vary with the solvent. Therefore, different references give slightly different values. Appendix 4 gives pK_a values that are consistent with the relative strengths of acids and bases, even if the values differ from one source to another.

PROBLEM 2-12

For each of the following reactions, suggest which solvent(s) would be compatible with the acids and bases involved. (We will ignore any other possible reactions for now.) Your choices of solvents are pentane, diethyl ether, ethanol, water, and ammonia. Refer to Appendix 4 for any needed values of pK_a , or estimate them.



PROBLEM-SOLVING HINT

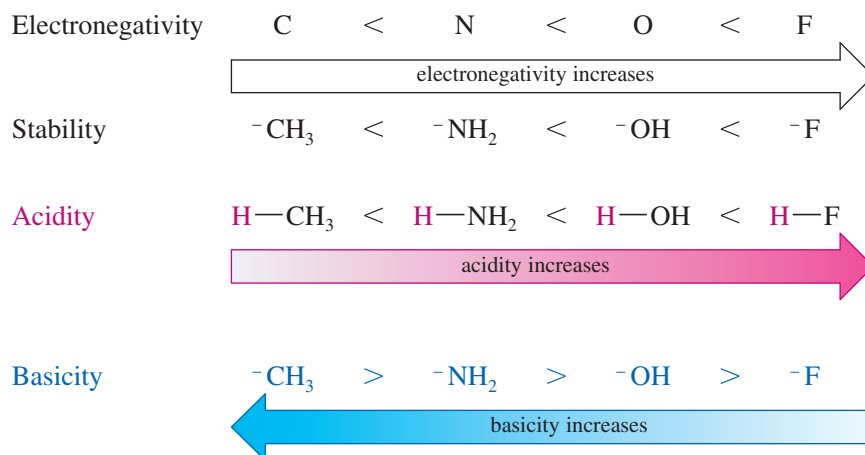
Water and alcohols ($R-O-H$) can react as both weak acids and weak bases, and they limit the strength of strong acids and bases dissolved in them. Ethers ($R-O-R$) can react as weak bases, and they limit the strength of strong acids (but not bases) dissolved in them. Alkanes do not react as either acids or bases, and they do not limit the strength of acids or bases dissolved in them.

2-9 Effects of Size and Electronegativity on Acidity

How can we look at a structure and predict whether a compound will be a strong acid, a weak acid, or not an acid at all? To be a Brønsted-Lowry acid (HA), a compound must contain a hydrogen atom that can be lost as a proton. A strong acid must have a stable conjugate base (A^-) after losing the proton.

The stability of the conjugate base is a good guide to acidity. More stable anions tend to be weaker bases, and their conjugate acids tend to be stronger acids. Some of the factors that affect the stability of conjugate bases are electronegativity, size, and resonance.

Electronegativity A more electronegative element bears a negative charge more easily, giving a more stable conjugate base and a stronger acid. Electronegativities increase from left to right in the periodic table:

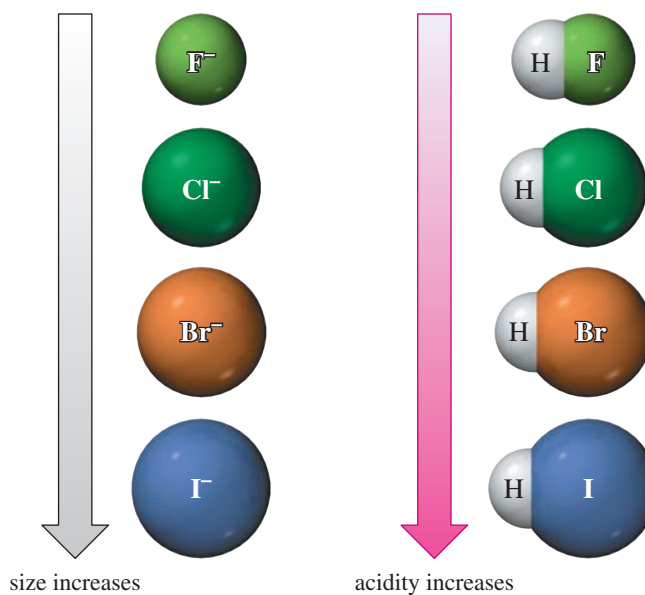


PROBLEM-SOLVING HINT

To evaluate or compare the acidity of acids, consider the stability of their conjugate bases.

To compare the basicity of bases, consider the stability of their conjugate acids.

Size The negative charge of an anion is more stable if it is spread over a larger region of space. Within a column of the periodic table, acidity increases down the column, as the size of the element increases.



PROBLEM-SOLVING HINT

In the comparison of similar bases, a negative charge is generally more stable on a larger atom or on a more electronegative atom. Enhanced stability of a base increases the strength of its conjugate acid.

Figure 2-11 summarizes these two effects, using part of the periodic table to show the $\text{p}K_{\text{a}}$ of the acid formed when each element is combined with enough hydrogen atoms to fill its valence. Acidity increases toward the right (increasing electronegativity) and downward (increasing size) on the periodic table. A smaller or more negative value of $\text{p}K_{\text{a}}$ implies a stronger acid and a more stable conjugate base.

IVA	VA	VIA	VIIA
CH ₄ pK _a 50	NH ₃ pK _a 36	H ₂ O pK _a 15.7	HF pK _a 3.2
	PH ₃ pK _a 29	H ₂ S pK _a 7.0	HCl pK _a -7
		H ₂ Se pK _a 3.9	HBr pK _a -9
		H ₂ Te pK _a 2.6	HI pK _a -10

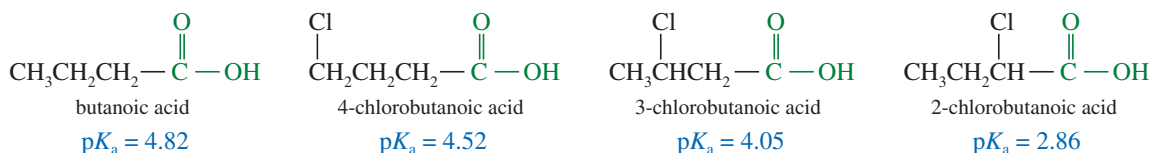
FIGURE 2-11 Acidity trends within the periodic table. Acidity depends on the stability of the conjugate base, which increases with the electronegativity and size of the anion. Electronegativity increases toward the right of the periodic table, and size increases down each column.

2-10 Inductive Effects on Acidity

As we have seen, the strength of an acid is determined by the stability of its conjugate base. Electron-withdrawing atoms and groups can stabilize a conjugate base through the sigma bonds of the molecule. Stabilization of the conjugate base results in a stronger acid (lower value of pK_a). The magnitude of this **inductive effect** depends on the number of bonds between the electronegative element (or other electron-withdrawing group) and the site of the negative charge. Note how adding a chlorine atom to butanoic acid increases its acidity, and the effect is larger if the chlorine atom is closer to the acidic group.

PROBLEM-SOLVING HINT

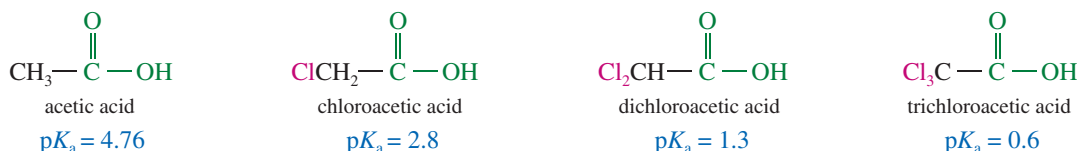
When a problem asks you to compare acidities or explain an acidity, you should consider the structure and stability of the conjugate base. Factors that stabilize the conjugate base will increase the strength of the acid.



Stronger electron-withdrawing groups stabilize the anion of the conjugate base more than weaker groups do, leading to stronger acids. Fluorine is more electronegative and a stronger withdrawing group than chlorine, making fluoroacetic acid a stronger acid than chloroacetic acid. This trend continues through the other halogen-substituted acetic acids.



Multiple electron-withdrawing groups stabilize the conjugate base and increase the acidity more than a single group does. The following examples show how each additional chlorine atom increases the acidity of the substituted acetic acid.



PROBLEM-SOLVING HINT

Acidity is increased by

1. stronger electron-withdrawing substituents.
2. additional electron-withdrawing substituents.
3. closer proximity of the electron-withdrawing substituent to the acidic group.

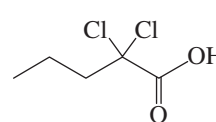
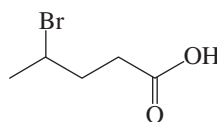
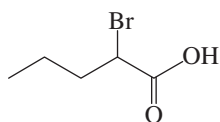
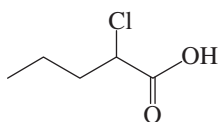
PROBLEM 2-13

Write equations for the following acid–base reactions. Label the conjugate acids and bases, and show any inductive stabilization. Predict whether the equilibrium favors the reactants or products. Try to do this without using a table of pK_a values, but if you need a hint, you can consult Appendix 4.

- (a) $\text{CH}_3\text{CH}_2\text{OH} + \text{CH}_3\text{NH}^-$ (b) $\text{F}_3\text{CCOONa} + \text{Br}_3\text{C}-\text{COOH}$
 (c) $\text{CH}_3\text{OH} + \text{H}_2\text{SO}_4$ (d) $\text{NaOH} + \text{H}_2\text{S}$
 (e) $\text{CH}_3\text{NH}_3^+ + \text{CH}_3\text{O}^-$ (f) $\text{BrCH}_2\text{CH}_2\text{OH} + \text{F}_3\text{C}-\text{CH}_2\text{O}^-$
 (g) $\text{NaOCH}_2\text{CH}_3 + \text{Cl}_2\text{CHCH}_2\text{OH}$ (h) $\text{H}_2\text{Se} + \text{NaNH}_2$
 (i) $\text{CH}_3\text{CHF}\text{COOH} + \text{FCH}_2\text{CH}_2\text{COO}^-$ (j) $\text{CF}_3\text{CH}_2\text{O}^- + \text{FCH}_2\text{CH}_2\text{OH}$

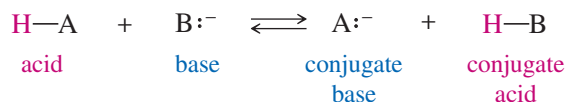
PROBLEM 2-14

Rank the following acids in decreasing order of their acid strength. In each case, explain why the previous compound should be a stronger acid than the one that follows it.

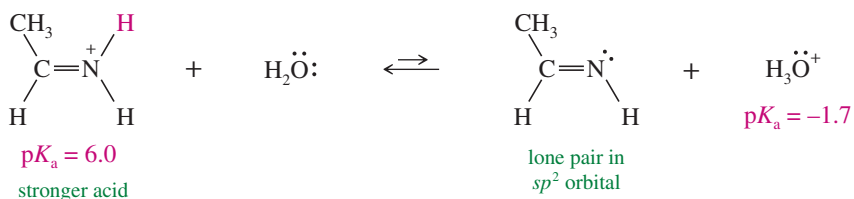
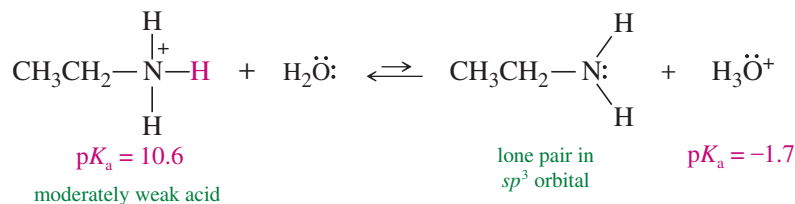


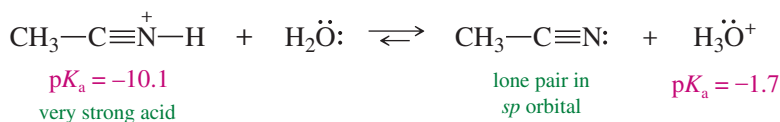
2-11 Hybridization Effects on Acidity

When an acid donates a proton, the electrons that were originally in the $\text{H}-\text{A}$ bond remain with the conjugate base (A^-) as a nonbonding electron pair (lone pair).

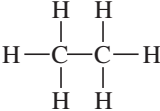
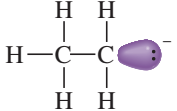
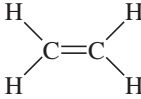
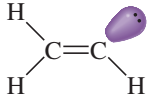


The stability of the conjugate base, and thus the strength of the acid, is determined by the stability of this nonbonding electron pair. As we have seen in Sections 2-9 and 2-10, the stability of this lone pair depends on the electronegativity, size, charge, and inductive substituents of the atom bearing it. The stability of the lone pair also depends on the hybridization state of the orbital where it resides. For example, consider the compounds below. All of them can lose a proton from a positively charged nitrogen atom with four bonds to give a neutral nitrogen with three bonds and a lone pair. Yet their different hybridization states strongly influence their acidity. Note that the acidity of the $\text{N}-\text{H}$ bond varies with hybridization in the order $sp^3 < sp^2 \ll sp$.



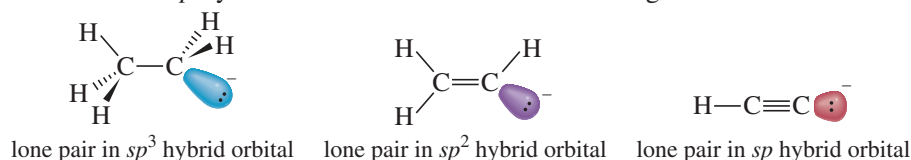


The analogous carbon compounds also show this hybridization dependence. The following carbon compounds are all very weak acids, yet their acidity varies in the same order as we saw with nitrogen: $sp^3 < sp^2 < sp$.

Compound	Conjugate Base	Hybridization	s Character	$\text{p}K_{\text{a}}$
		sp^3	25%	50
		sp^2	33%	44
		sp	50%	25

We explain this hybridization dependence by considering the relative stability of the s and p components of these hybrid orbitals. In Section 1-2B, we saw that an electron in the $2s$ orbital is lower in energy than one in the $2p$ orbital. Therefore, a nonbonding pair of electrons remaining in a hybrid orbital (after an acid has donated a proton) is more stable in a hybrid using more of the s orbital and less of the p orbitals. The amount of the s orbital (s character) used in each hybrid orbital is 25% in the sp^3 hybrid, 33% in the sp^2 hybrid, and 50% in the sp hybrid. Thus, both the s character of the hybrid orbital, and the stability of a lone pair in the orbital, increase in the order $sp^3 < sp^2 < sp$.

The shapes of the orbitals also reflect how increasing the s character results in the electrons being held closer to the nucleus. The drawings below show lone pairs in sp^3 , sp^2 , and sp hybrid orbitals. Electrons in the sp^3 hybrid orbital are, on average, farthest from the nucleus. Electrons in the sp^2 hybrid orbital are held closer and more tightly. Electrons in the sp hybrid orbital are held the closest and tightest of all.



PROBLEM 2-15

Like nitrogen and carbon, oxygen also shows this same hybridization effect on acidity. Both of the following compounds can lose a proton from a positively charged oxygen with three bonds to give a conjugate base containing a neutral oxygen with two bonds. One of these structures has $\text{p}K_{\text{a}} = -2.4$, while the other has $\text{p}K_{\text{a}} = -8.0$.

- (a) Show the reaction of each compound with water.
 (b) Match each structure with its $\text{p}K_{\text{a}}$, and explain your choice.

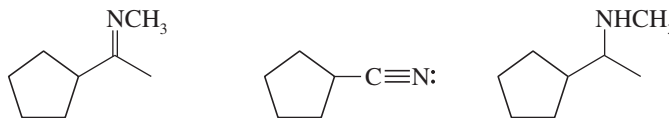


PROBLEM-SOLVING HINT

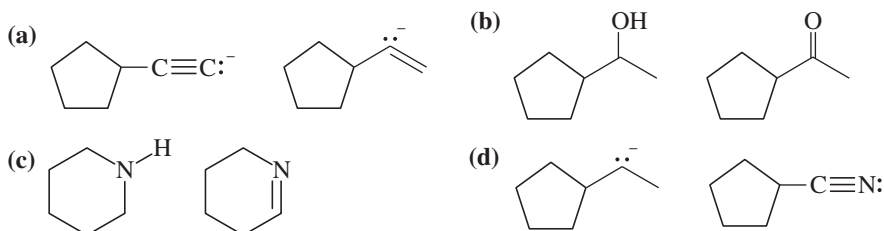
If we compare two similar acids that donate a proton from the same element with the same charge, differing only in the hybridization of the atom, acidity increases in the order $sp^3 < sp^2 < sp$.

PROBLEM 2-16

Consider the type of orbitals involved, and rank the following nitrogen compounds in order of *decreasing basicity*. Give the structures of their conjugate acids, and estimate their pK_a s from similar compounds in Appendix 4. Rank the conjugate acids in order of *increasing acidity*. (Hint: These two orders should be the same!)

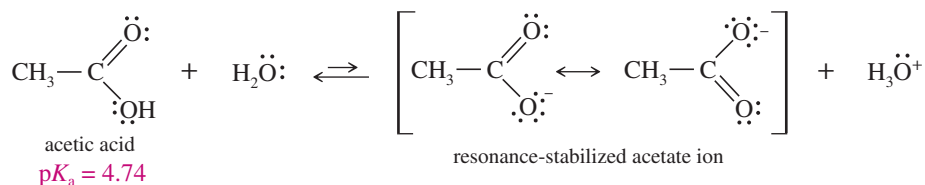
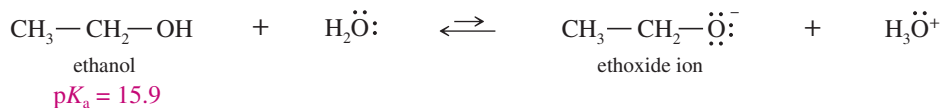
**PROBLEM 2-17**

Consider each pair of bases, and explain which one is more basic. Draw their conjugate acids, and show which one is a stronger acid.



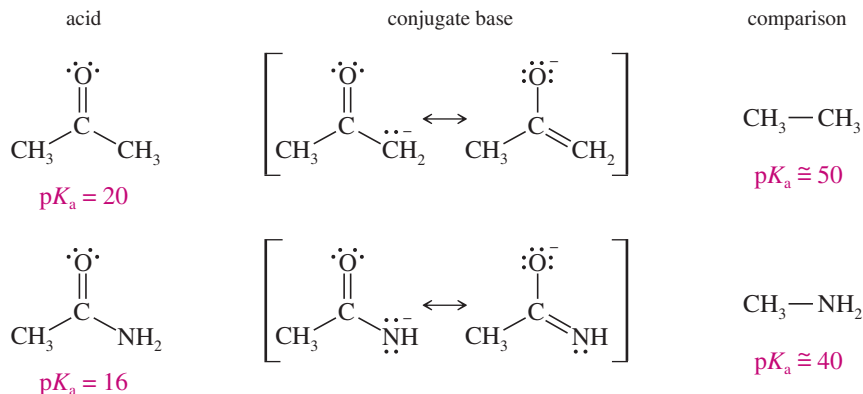
2-12 Resonance Effects on Acidity and Basicity

When an acid donates a proton, the conjugate base is left with the two electrons that once bonded it to the proton, and often a negative charge. This lone pair of electrons and the negative charge (if any) may be delocalized over two or more atoms by resonance. The amount of stabilization that occurs through resonance depends on how electronegative the atoms are and how many share the charge and the lone pair. In many cases, resonance delocalization is the dominant effect helping to stabilize the conjugate base. For example, acetic acid is over 11 pK_a units more acidic than ethanol (see below). This large difference is due to enhanced stability of the acetate ion, which is the conjugate base of acetic acid. The negative charge in the acetate ion is delocalized over two oxygen atoms, in contrast to the ethoxide ion, which has the negative charge localized on its single oxygen atom.

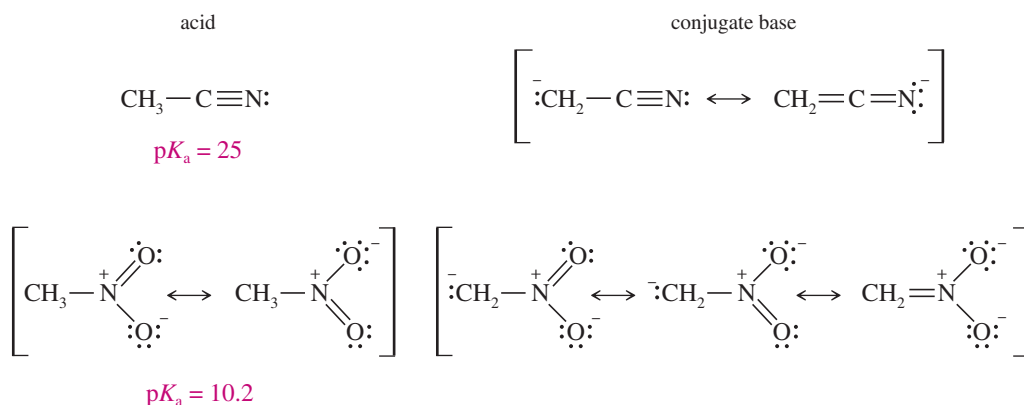
**PROBLEM 2-18**

Which is a stronger base: cyanide ion or fluoride ion? Give pK_b values (without looking them up) to support your choice.

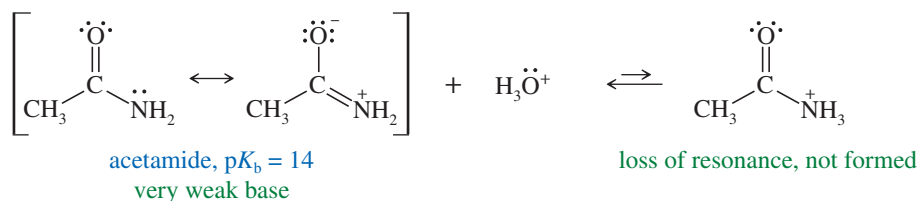
Protons on carbon and nitrogen are also more acidic if they are located where the conjugate base is stabilized by resonance. The following examples show how resonance stabilization of the conjugate base can enhance their acidity. Each of these examples has a carbonyl ($\text{C}=\text{O}$) group next to the atom that donates the proton. These are both weak acids, but without the carbonyl group, they would be much weaker.



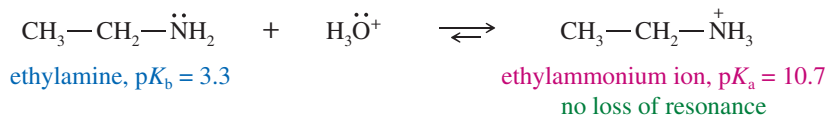
Other electron-withdrawing groups can also stabilize a conjugate base through resonance. Two of the strongest stabilizing groups are the cyano ($-\text{C}\equiv\text{N}$) and nitro ($-\text{NO}_2$) groups.



Resonance stabilization also affects the basicity of common organic compounds. This effect is seen in amides, which are much less basic than amines. An amide is stabilized by resonance between the amino ($-\text{NH}_2$) group and the carbonyl ($\text{C}=\text{O}$) group. This resonance delocalizes the lone pair on nitrogen and makes it less available for bonding to a proton. It also gives the nitrogen atom a partial positive charge. If the nitrogen atom were to protonate as shown below, the resonance stabilization would be lost.



Compare this amide's very weak basicity with the corresponding amine, which is over 10 pK_b units more basic.



Resonance delocalization is the most common type of stabilization we see in the conjugate bases of common organic acids. Resonance is often the largest stabilizing factor as well, sometimes contributing 10 or more pK_a units to the acidity of the conjugate acid.

PROBLEM-SOLVING HINT

Resonance delocalization of the charge(s) on a base helps to stabilize the base, making it a weaker base and strengthening its conjugate acid. The strongest common stabilizing groups are the carbonyl ($\text{C}=\text{O}$), cyano ($\text{C}\equiv\text{N}$), and nitro (NO_2) groups.

PROBLEM 2-19

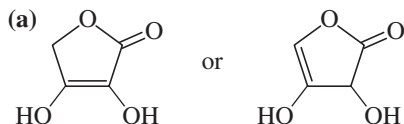
The preceding equation for the protonation of acetamide shows a hypothetical product that is not actually formed. When acetamide is protonated by a strong enough acid, it does not protonate on nitrogen, but at a different basic site. Draw the structure of the actual conjugate acid of acetamide, and explain (resonance) why protonation occurs where it does rather than on nitrogen. Calculate the pK_a of this conjugate acid.

PROBLEM 2-20

Acetic acid can also react as a very weak base ($pK_b = 20$). Two different sites on acetic acid might become protonated to give the conjugate acid. Draw both of these possible conjugate acids, and explain (resonance) why the correct one is more stable. Calculate the pK_a of this conjugate acid.

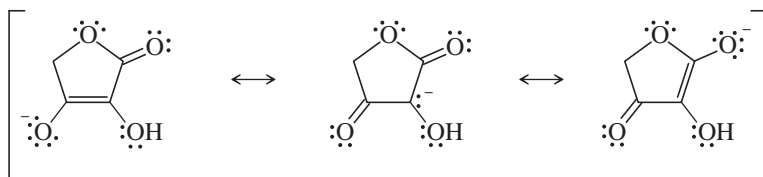
PROBLEM 2-21 (PARTIALLY SOLVED)

Choose the more acidic member of each pair of isomers, and show why the acid you chose is more acidic.

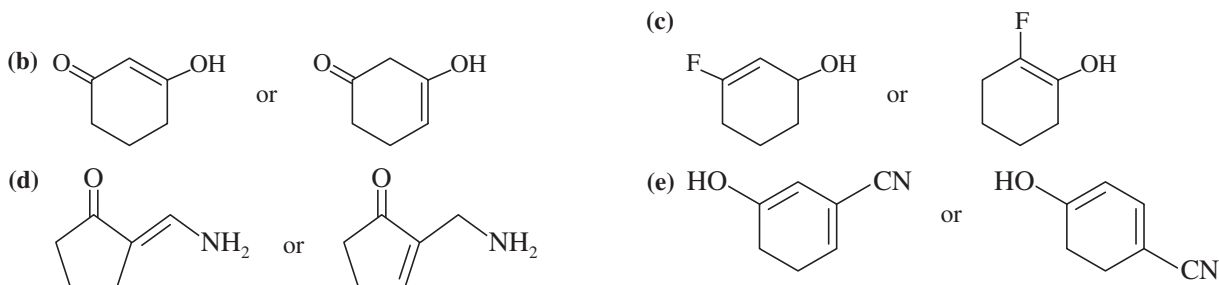


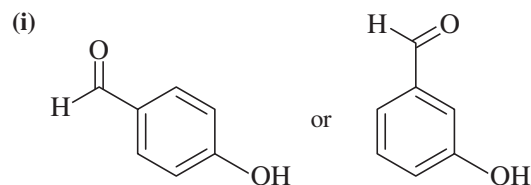
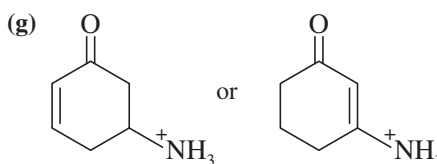
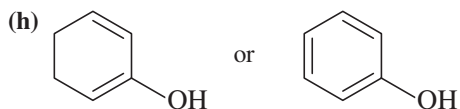
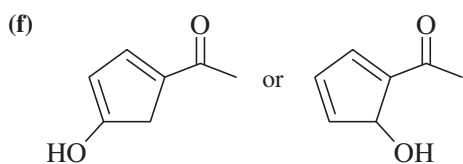
SOLUTION

to (a): The first (left) compound is more acidic because deprotonation of its left —OH group gives a resonance-stabilized anion that delocalizes the negative charge onto the carbonyl group. Compare this compound with vitamin C, shown at the beginning of the chapter. This resonance explains why vitamin C (ascorbic acid) is acidic even though it does not contain a carboxylic acid (—COOH) group.



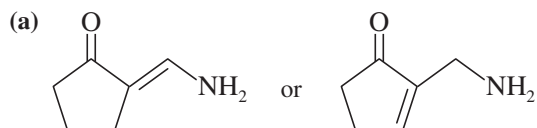
In the second compound, deprotonation of either —OH group fails to give an anion that delocalizes the negative charge onto the carbonyl group.





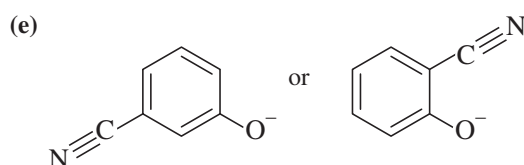
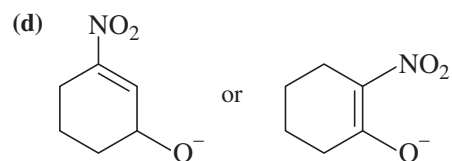
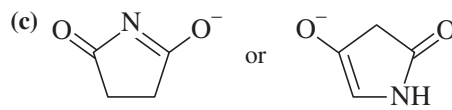
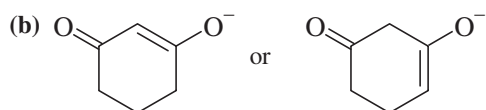
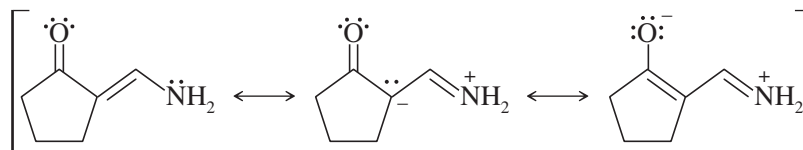
PROBLEM 2-22 (PARTIALLY SOLVED)

Choose the more basic member of each pair of isomers, and show why the base you chose is more basic.



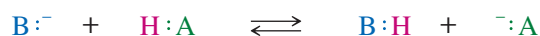
SOLUTION

to (a): The first (left) compound is less basic because it is resonance-stabilized (like an amide), and protonation of its —NH_2 group would destroy that resonance. Moreover, the lone pair on the —NH_2 group is shared with the carbonyl group, and the —NH_2 group carries a partial positive charge. The compound on the right is an amine, which we expect to be more basic. Protonation of its —NH_2 group would not cause loss of resonance stabilization.



2-13 Lewis Acids and Bases

The Brønsted–Lowry definition of acids and bases depends on the transfer of a proton from the acid to the base. The base uses a pair of nonbonding electrons to form a bond to the proton.

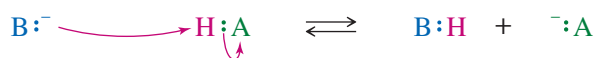


In 1923, Gilbert N. Lewis (University of California, Berkeley) reasoned that this kind of reaction does not need a proton. Instead, a base could use its lone pair of electrons to bond to some other electron-deficient atom. In effect, we can look at an acid–base reaction from the viewpoint of the *bonds* that are formed and broken rather

PROBLEM-SOLVING HINT

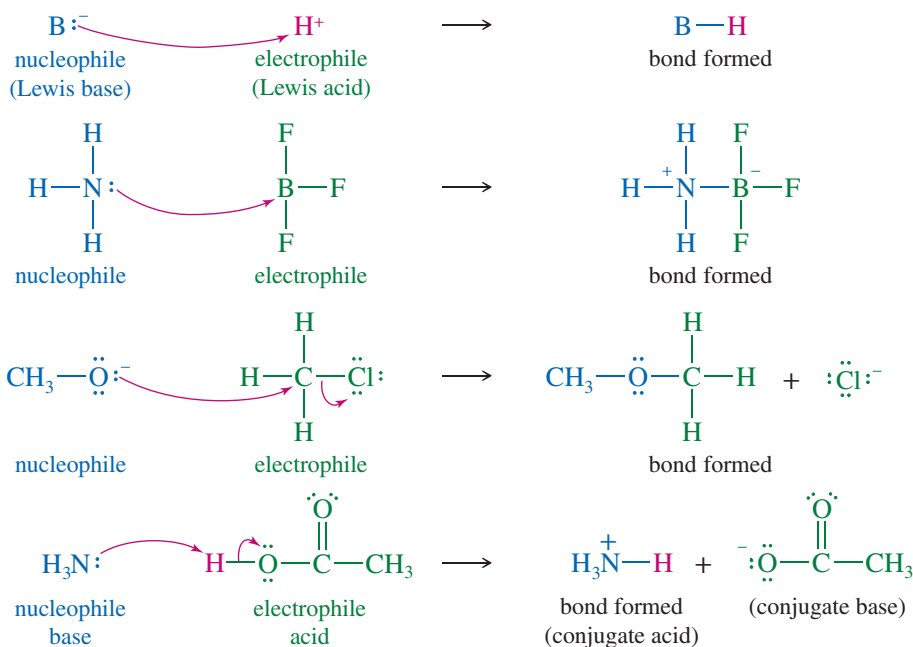
A nucleophile donates electrons.
An electrophile accepts electrons.
Acidic protons may serve as electron acceptors.

than a proton that is transferred. The following reaction shows the proton transfer, with emphasis on the bonds being broken and formed. Organic chemists routinely use curved arrows to show the movement of the participating electrons.



Lewis bases are species with available electrons that can be donated to form new bonds. **Lewis acids** are species that can accept these electron pairs to form new bonds. Because a Lewis acid *accepts* a pair of electrons, it is called an **electrophile**, from the Greek words meaning “lover of electrons.” A Lewis base is called a **nucleophile**, or “lover of nuclei,” because it donates electrons to a nucleus with an empty (or easily vacated) orbital. In this book, we sometimes use colored type for emphasis: blue for nucleophiles, green for electrophiles, and occasionally red for acidic protons.

The Lewis acid–base definitions include reactions having nothing to do with protons. Following are some examples of Lewis acid–base reactions. Notice that the common Brønsted–Lowry acids and bases also fall under the Lewis definition, with a proton serving as the electrophile. Curved arrows (red) are used to show the movement of electrons, generally from the nucleophile to the electrophile.

**PROBLEM-SOLVING HINT**

So far, we have used several types of arrows in this course.

Reaction arrows: $\longrightarrow$

Equilibrium arrows: $\rightleftharpoons$

Resonance arrows: $\longleftrightarrow$

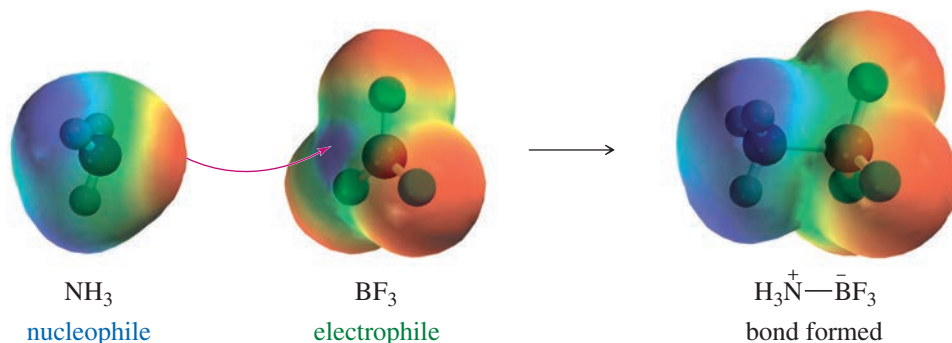
Flow of electrons in

Lewis acid–base reactions:

Imaginary flow of electrons between resonance forms:

Some of the terms associated with acids and bases have evolved specific meanings in organic chemistry. When organic chemists use the term *base*, they usually mean a proton acceptor (a Brønsted–Lowry base). Similarly, the term *acid* usually means a proton donor (a Brønsted–Lowry acid). When the acid–base reaction involves formation of a bond to some other element (especially carbon), organic chemists refer to the electron donor as a *nucleophile* (Lewis base) and the electron acceptor as an *electrophile* (Lewis acid).

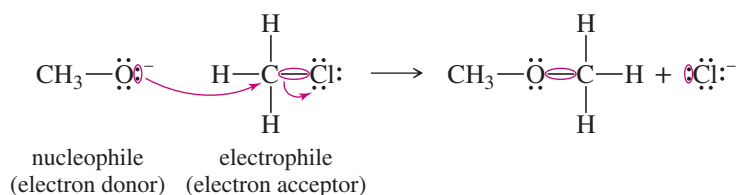
The following illustration shows electrostatic potential maps for the reaction of NH_3 (the nucleophile/electron donor) with BF_3 (the electrophile/electron acceptor). The electron-rich (red) region of NH_3 attacks the electron-poor (blue) region of BF_3 . In contrast, the product shows high electron density on the boron atom and its three fluorine atoms and low electron density on nitrogen and its three hydrogen atoms.



2-14 The Curved-Arrow Formalism

We often break down complicated organic reactions and biochemical reactions into series of steps involving simple Lewis acid–base reactions of nucleophiles with electrophiles. This section describes the mechanics of drawing these individual steps using curved arrows to show movements of the electrons.

The **curved-arrow formalism** is used to show the flow of an electron pair *from the electron donor to the electron acceptor*. The movement of each pair of electrons involved in making or breaking bonds is indicated by its own separate arrow, as shown in the preceding set of reactions. In this book, these curved arrows are always printed in red. In the preceding reaction of CH_3O^- with CH_3Cl , one curved arrow shows the lone pair on oxygen forming a bond to carbon. Another curved arrow shows that the $\text{C} - \text{Cl}$ bonding pair detaches from carbon and becomes a lone pair on the Cl^- product.



PROBLEM-SOLVING HINT

Use one curved arrow for each pair of electrons participating in the reaction.

The curved-arrow formalism is universally used for keeping track of the flow of electrons in reactions. In this way, we show exactly what changes take place in a reaction, as well as the details of how those changes occur. We have also used this device (in Section 1-9, for example) to keep track of electrons in resonance structures as we imagined their “flow” in going from one resonance structure to another. We colored them green in that case, rather than red, because electrons do not actually “flow” in resonance structures; they are simply delocalized. Still, the curved-arrow formalism helps our *minds* flow from one resonance structure to another. We will find ourselves constantly using these curved arrows to keep track of electrons, both as reactants change to products and as we imagine additional resonance structures of a hybrid.

PROBLEM 2-23 (PARTIALLY SOLVED)

In the following acid–base reactions,

1. draw Lewis structures of the reactants and the products.
2. determine which species are acting as electrophiles (acids) and which are acting as nucleophiles (bases).
3. use the curved-arrow formalism to show the movement of electron pairs in these reactions, as well as the imaginary movement in the resonance hybrids of the products.

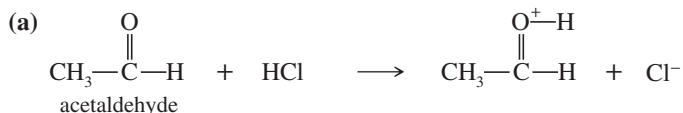
(continued)

PROBLEM-SOLVING HINT

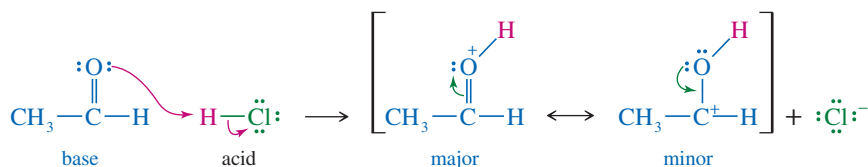
The curved red arrows show movement of electrons. Each arrow must begin at a pair of electrons: either an unshared electron pair or a pair of electrons in a bond.

Each arrow should point toward where that pair of electrons will be in the products, either forming a bond to another atom or residing as an unshared pair on an atom.

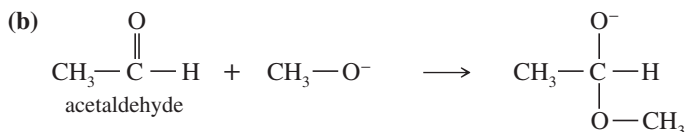
4. indicate which reactions are best termed Brønsted–Lowry acid–base reactions.



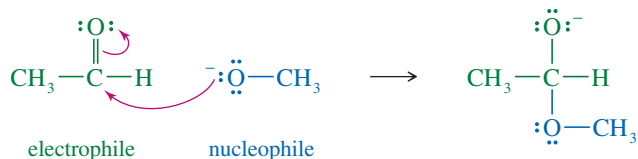
This reaction is a proton transfer from HCl to the C=O group of acetaldehyde. Therefore, it is a Brønsted–Lowry acid–base reaction, with HCl acting as the acid (proton donor) and acetaldehyde acting as the base (proton acceptor). Before drawing any curved arrows, remember that one arrow must show the movement of electrons *from* the electron-pair donor (the nucleophile/base) *to* the electron-pair acceptor (the electrophile/acid). Another curved arrow must show the electrons that once formed the H—Cl bond leaving with chlorine to give a chloride ion. Drawing these arrows is easier once we draw valid Lewis structures for all the reactants and products.



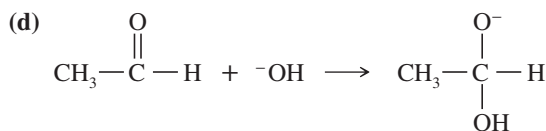
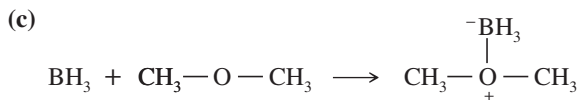
The resonance forms of the product show that a pair of electrons can be moved between the oxygen atom and the C=O pi bond. The positive charge is delocalized over the carbon and oxygen atoms, with most of the positive charge on oxygen because all octets are satisfied in that resonance structure.

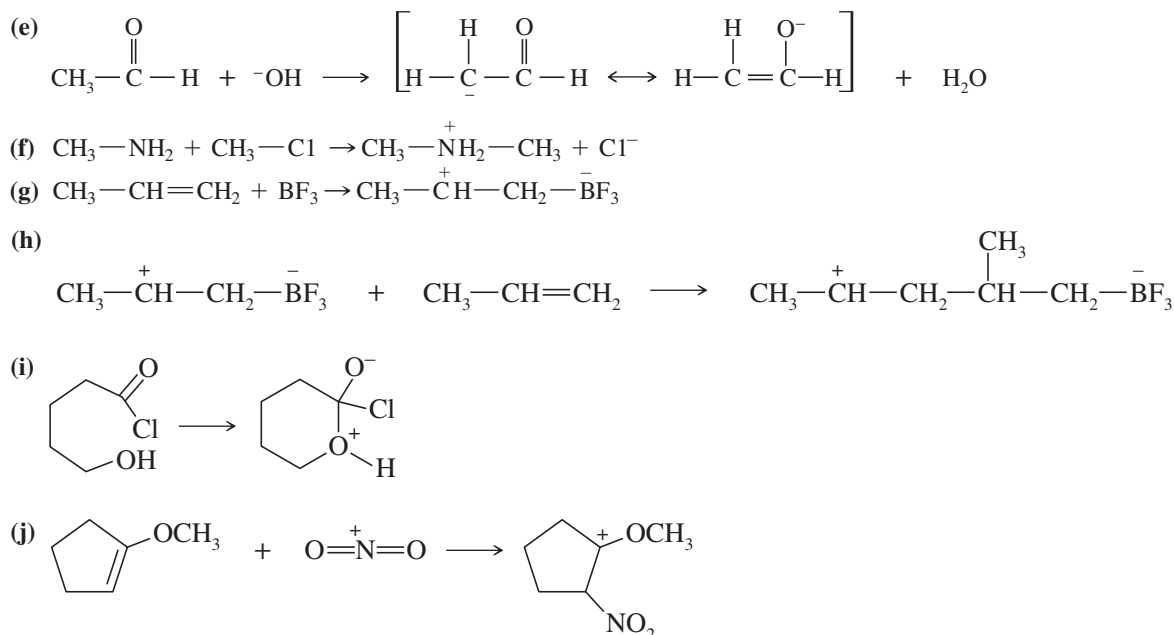


In this case, no proton has been transferred, so this is not a Brønsted–Lowry acid–base reaction. Instead, a bond has formed between the oxygen of the CH₃—O[−] group and the C=O carbon atom. Drawing the Lewis structures helps to show that the CH₃—O[−] group (the nucleophile in this reaction) donates the electrons to form the new bond to acetaldehyde (the electrophile). This result agrees with our intuition that a negatively charged ion is likely to be electron-rich and therefore an electron donor.



Notice that acetaldehyde acts as the nucleophile (Lewis base) in part (a) and as the electrophile (Lewis acid) in part (b). Like most organic compounds, acetaldehyde is both acidic and basic. It acts as a base if we add a strong enough acid to make it donate electrons or accept a proton. It acts as an acid if the base we add is strong enough to donate an electron pair or abstract a proton.





2-15 Hydrocarbons

In later chapters, we will study many different types of organic compounds. The various kinds of compounds are briefly described here so that you can recognize them as you encounter them. For the purpose of this brief survey, we divide organic compounds into three classes: (1) hydrocarbons, (2) compounds containing oxygen, and (3) compounds containing nitrogen.

The **hydrocarbons** are compounds composed entirely of carbon and hydrogen. The major classes of hydrocarbons are alkanes, alkenes, alkynes, and aromatic hydrocarbons. Nearly all hydrocarbons are nonpolar or only slightly polar. Therefore, hydrocarbons (and hydrocarbon parts of larger molecules) tend to be hydrophobic.

2-15A Alkanes

Alkanes are hydrocarbons that contain only single bonds. Alkane names generally have the *-ane* suffix, and the first part of the name indicates the number of carbon atoms. Table 2-3 shows how the prefixes in the names correspond with the number of carbon atoms. Naming of compounds (**nomenclature**) is covered in more detail in the chapters devoted to the individual functional groups and summarized in Appendix 5 (*Summary of Organic Nomenclature*).

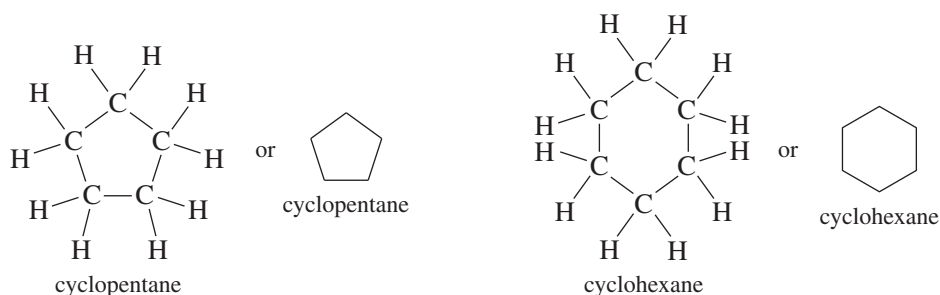
TABLE 2-3 Correspondence of Prefixes and Numbers of Carbon Atoms

Alkane Name	Number of Carbons	Alkane Name	Number of Carbons
methane	1	hexane	6
ethane	2	heptane	7
propane	3	octane	8
butane	4	nonane	9
pentane	5	decane	10

CH_4	CH_3-CH_3	$\text{CH}_3-\text{CH}_2-\text{CH}_3$	$\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_3$	$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3-\text{CH}-\text{CH}_3 \end{array}$
methane	ethane	propane	butane	isobutane

The **cycloalkanes** are a special class of alkanes in the form of a ring. Figure 2-12 shows the Lewis structures and line-angle formulas of cyclopentane and cyclohexane, the cycloalkanes containing five and six carbons, respectively.

FIGURE 2-12 Cycloalkanes. Cycloalkanes are alkanes in the form of a ring.



Alkanes are the major components of heating gases (natural gas and liquefied petroleum gas), gasoline, jet fuel, diesel fuel, motor oil, fuel oil, and paraffin “wax.” Other than combustion, alkanes undergo few reactions. In fact, when a molecule contains an alkane portion and a nonalkane portion, we often ignore the presence of the alkane portion because it is relatively unreactive. Alkanes undergo few reactions because they have no **functional group**, the part of the molecule where reactions usually occur. Functional groups are distinct chemical units, such as double bonds, hydroxy groups, or halogen atoms, that are reactive. Most organic compounds are characterized and classified by their functional groups.

An **alkyl group** is an alkane portion of a molecule, with one hydrogen atom removed to allow bonding to the rest of the molecule. Figure 2-13 shows an ethyl group (C_2H_5) attached to cyclohexane to give ethylcyclohexane. We might try to name this compound as “cyclohexylethane,” but we should treat the larger fragment as the parent compound (cyclohexane) and the smaller group as the alkyl group (ethyl).

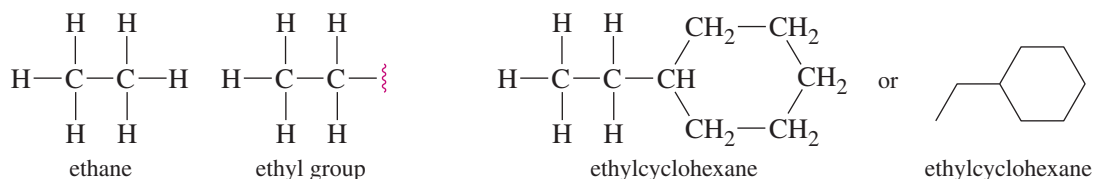
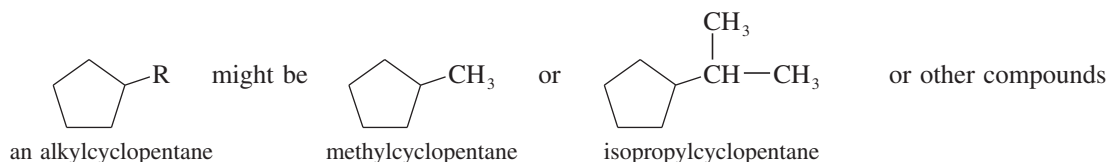


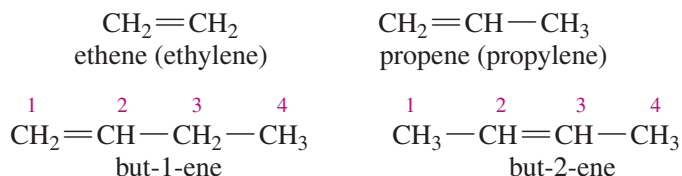
FIGURE 2-13 Naming alkyl groups. Alkyl groups are named like the alkanes they are derived from, with a -yl suffix.

We are often concerned primarily with the structure of the most important part of a molecule. In these cases, we can use the symbol R as a substituent to represent an alkyl group (or some other unreactive group). We presume that the exact nature of the R group is unimportant.

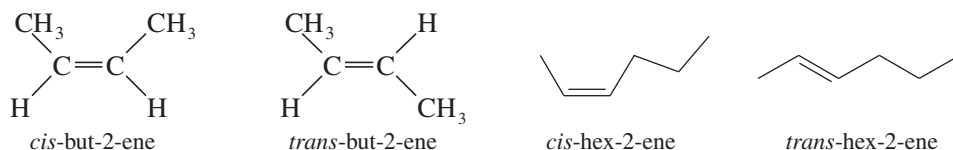


2-15B Alkenes

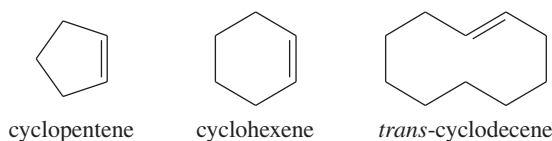
Alkenes are hydrocarbons that contain carbon–carbon double bonds. A carbon–carbon double bond is the most reactive part of an alkene, so we say that the double bond is the *functional group* of the alkene. Alkene names end in the *-ene* suffix. If the double bond might be in more than one position, then the chain is numbered and the lower number of the two double-bonded carbons is added to the name to indicate the position of the double bond.



Carbon–carbon double bonds cannot rotate, and many alkenes show geometric (cis-trans) isomerism (Sections 1-18B and 1-19B). Following are the cis-trans isomers of some simple alkenes:

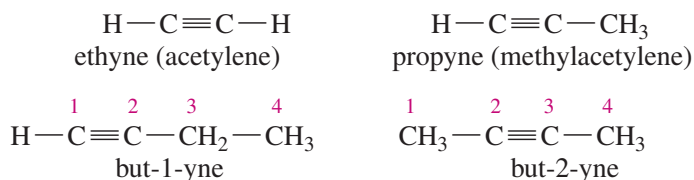


Cycloalkenes are also common. Unless the rings are very large, cycloalkenes are always the cis isomers, and the term *cis* is omitted from the names. In a large ring, a trans double bond may occur, giving a *trans*-cycloalkene.

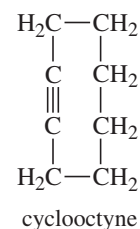


2-15C Alkynes

Alkynes are hydrocarbons with carbon–carbon triple bonds as their functional group. Alkyne names generally have the *-yne* suffix, although some of their common names (*acetylene*, for example) do not conform to this rule. The triple bond is linear, so there is no possibility of geometric (cis-trans) isomerism in alkynes.



In an alkyne, four atoms must be in a straight line. These four collinear atoms are not easily bent into a ring, so cycloalkynes are rare. Cycloalkynes are stable only if the ring is large, containing eight or more carbon atoms. See the structure of cyclooctyne shown in the margin.

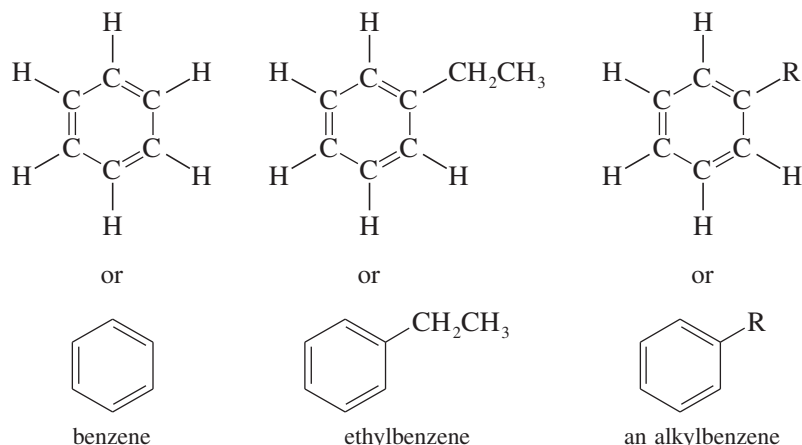


2-15D Aromatic Hydrocarbons

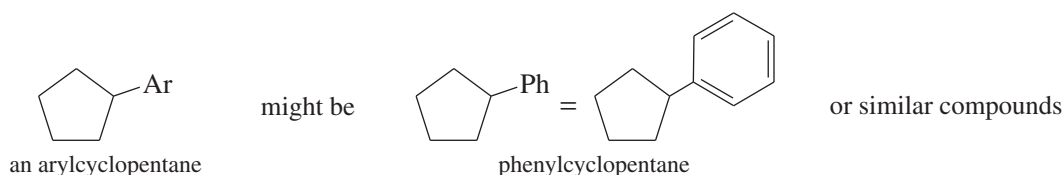
The following compounds may look like cycloalkenes, but their properties are different from those of simple alkenes. These **aromatic hydrocarbons** (also called **arenes**) are all derivatives of *benzene*, represented by a six-membered ring with three double bonds. This bonding arrangement is particularly stable, for reasons that we explain in Chapter 16.

Application: Toxicity

The presence of a methyl or ethyl group can make a big difference in biological systems. For example, benzene is quite toxic and causes leukemia, but methyl benzene (and ethyl benzene) are less toxic because enzymes can oxidize the methyl or ethyl group.



Just as a generic alkyl group substituent is represented by R, a generic aryl group is represented by Ar. When a benzene ring serves as a substituent, it is called a phenyl group, abbreviated Ph.

**PROBLEM 2-24**

Classify the following hydrocarbons, and draw a Lewis structure for each one. A compound may fit into more than one of the following classifications:

alkane
alkene
alkyne

cycloalkane
cycloalkene
cycloalkyne

aromatic hydrocarbon

(a) $(\text{CH}_3\text{CH}_2)_2\text{CHCH}(\text{CH}_3)_2$ (b) $\text{CH}_3\text{CHCHCH}_2\text{CH}_3$ (c) $\text{CH}_3\text{CCCH}_2\text{CH}_2\text{CH}_3$

(d) (e) (f)

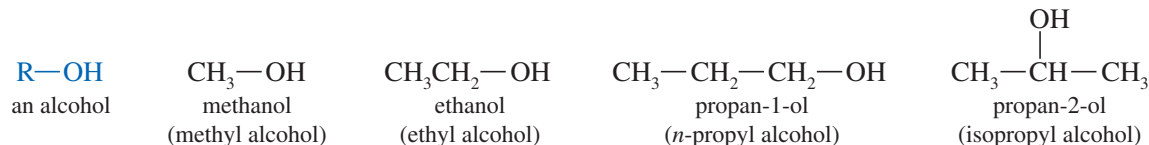
(g) (h) (i)

2-16 Functional Groups with Oxygen

Many organic compounds contain oxygen atoms bonded to alkyl groups. The major classes of oxygen-containing compounds are alcohols, ethers, ketones, aldehydes, carboxylic acids, and acid derivatives. We will cover their nomenclature in greater detail in upcoming chapters.

2-16A Alcohols

Alcohols are organic compounds that contain the **hydroxy group** (—OH) as their functional group. The general formula for an alcohol is R—OH . Alcohols are among the most polar organic compounds because the hydroxy group is strongly polar and can participate in hydrogen bonding. Thus, the hydroxy group is hydrophilic, and it confers solubility in water if the hydrophobic alkyl part of the molecule is not too large. Some of the simple alcohols such as ethanol and methanol are miscible (soluble in all proportions) with water, but alcohols with more than four carbons have limited solubility in water. Names of alcohols end in the *-ol* suffix from the word *alcohol*, as shown for the following common alcohols:



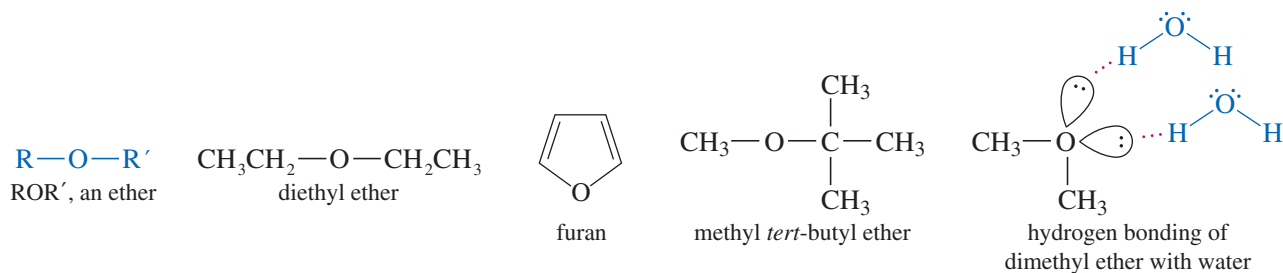
Alcohols are some of the most common organic compounds. Methyl alcohol (methanol), also known as “wood alcohol,” is used as an industrial solvent and as an automotive racing fuel. Ethyl alcohol (ethanol) is sometimes called “grain alcohol” because it is produced by the fermentation of grain or almost any other organic material. “Isopropyl alcohol” is the common name for propan-2-ol, used as “rubbing alcohol.”

2-16B Ethers

Ethers are composed of two alkyl groups bonded to an oxygen atom. The general formula for an ether is R—O—R' . (The symbol R' represents another alkyl group, either the same as or different from the first.) Like alcohols, ethers are much more polar than hydrocarbons, but ethers have no O—H hydrogens, so they cannot hydrogen bond with themselves. Ethers do form hydrogen bonds with hydrogen-bond donors such as alcohols, amines, and water, enhancing their solubility with these compounds. Ether names are often formed from the names of the alkyl groups and the word *ether*. Diethyl ether is the common “ether” used for starting engines in cold weather and once used for surgical anesthesia.

Application: Environment

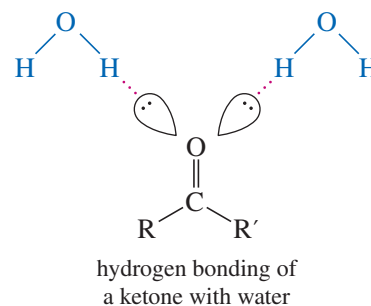
Ethanol is widely used as an additive to gasoline. “Gasohol” is an “oxygenated” fuel that contains about 10% ethanol, which causes the fuel to burn at a slightly lower temperature, producing lower concentrations of pollutants such as nitrogen oxides. Ethanol has replaced MTBE (methyl *tert*-butyl ether) as an oxygenated fuel additive because of safety concerns about MTBE contaminating groundwater when it leaks from storage tanks.



2-16C Aldehydes and Ketones

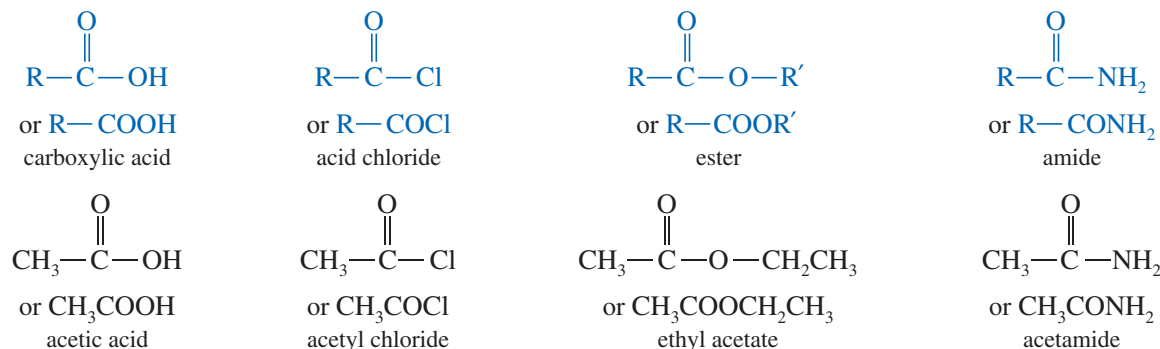
The **carbonyl group**, C=O , is the functional group for both aldehydes and ketones. A **ketone** has two alkyl groups bonded to the carbonyl group; an **aldehyde** has one alkyl group and a hydrogen atom bonded to the carbonyl group. Ketone names generally have the *-one* suffix; aldehyde names use either the *-al* suffix or the *-aldehyde* suffix.

The carbonyl group is strongly polar, and it can form hydrogen bonds with hydrogen-bond donors such as water, alcohols, and amines. The hydrophilic carbonyl group allows aldehydes and ketones containing up to four carbon atoms to be miscible with water. Those with more than four carbons have limited solubility in water.



2-16E Carboxylic Acid Derivatives

Carboxylic acids are easily converted to a variety of acid derivatives. Each derivative contains the carbonyl group bonded to an oxygen or another electron-withdrawing element. Among these functional groups are **acid chlorides**, **esters**, and **amides**. All of these groups can be converted back to carboxylic acids by acidic or basic hydrolysis.

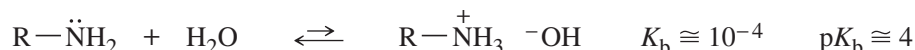


2-17 Functional Groups with Nitrogen

Nitrogen is another element often found in the functional groups of organic compounds. The most common “nitrogenous” organic compounds are amines, amides, and nitriles.

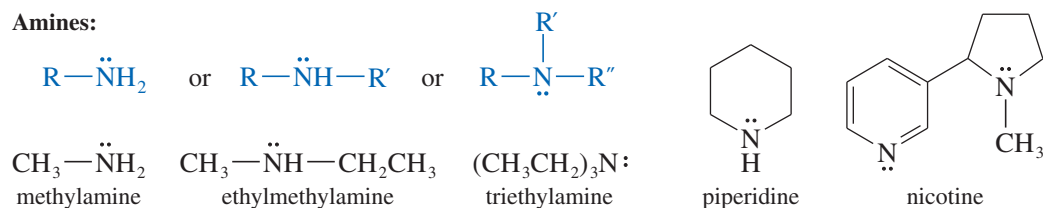
2-17A Amines

Amines are alkylated derivatives of ammonia. Like ammonia, amines are basic.



Because of their basicity (“alkalinity”), naturally occurring amines are often called *alkaloids*. Simple amines are named by naming the alkyl groups bonded to nitrogen and adding the word “amine.” The structures of some simple amines are shown below, together with the structure of nicotine, a toxic alkaloid found in tobacco leaves.

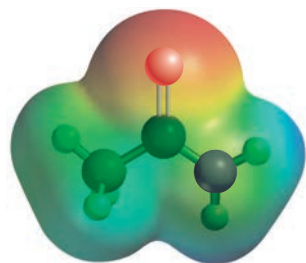
Amines:



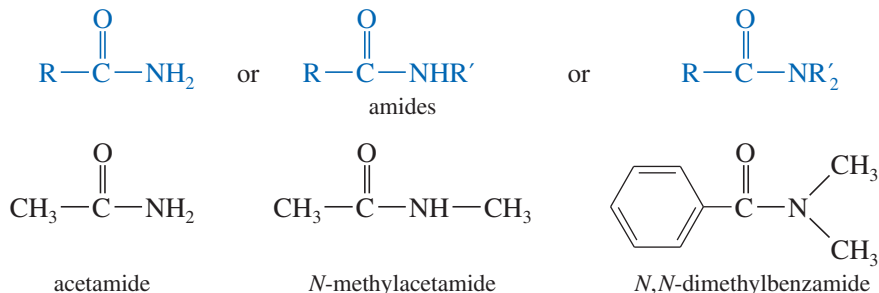
The nitrogen atom of an amine is hydrophilic because it can accept hydrogen bonds from water. If the amine has N—H bonds, these can donate hydrogen bonds to water. Therefore, many of the amines containing up to five carbon atoms are miscible with water.

2-17B Amides

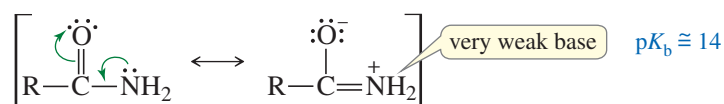
Amides are acid derivatives that result from a combination of an acid with ammonia or an amine. Proteins have the structure of long-chain, complex amides.



EPM of acetamide



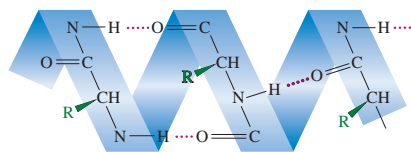
Amides are among the most stable acid derivatives. The nitrogen atom of an amide is not as basic as the nitrogen of an amine because of the electron-withdrawing effect of the carbonyl group (see Section 2-12). The following resonance forms help to show why amides are very weak bases:



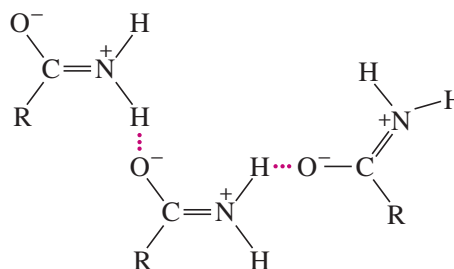
Application: Proteins

Proteins are specialized polymers of amides (Chapter 24). Their three-dimensional structures are defined and stabilized by the strong hydrogen bonding found in amides.

The alpha-helical structure shown below is held in position by hydrogen bonding between amide N—H groups and carbonyl groups in adjacent turns of the helix.



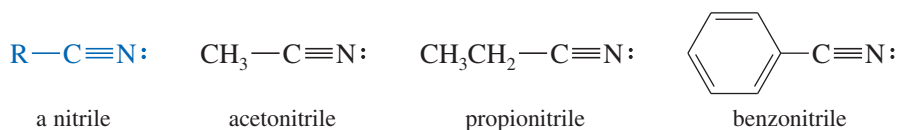
Amides form particularly strong hydrogen bonds, giving them high melting points and high boiling points. The strongly polarized amide N—H hydrogen forms unusually strong hydrogen bonds with the carbonyl oxygen that carries a partial negative charge in the polarized resonance form shown above. The following illustration shows this strong intermolecular hydrogen bonding.



hydrogen bonding in amides

2-17C Nitriles

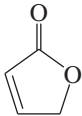
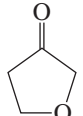
A **nitrile** is a compound containing the **cyano group**, $\text{—C}\equiv\text{N}$. The cyano group was introduced in Section 1-17 as an example of sp hybridized bonding. The cyano group is strongly polar by virtue of the $\text{C}\equiv\text{N}$ triple bond, and most small nitriles are somewhat soluble in water. Acetonitrile is miscible with water.

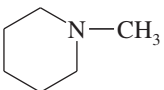
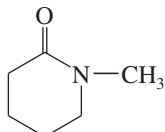
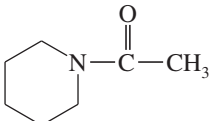
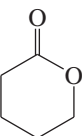
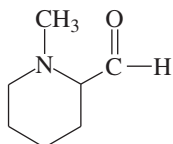
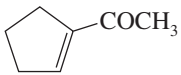


All of these classes of compounds are summarized in the table of Common Organic Compounds and Functional Groups, given on the front inside cover for convenient reference.

PROBLEM 2-26

Draw a Lewis structure, and classify each of the following compounds:

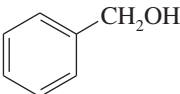
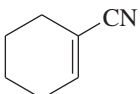
- (a) $\text{CH}_3\text{CH}_2\text{CONHCH}_3$ (b) $(\text{CH}_3\text{CH}_2)_2\text{NH}$ (c) $(\text{CH}_3)_2\text{CHCOOCH}_3$
 (d) $\text{CH}_3\text{CHCHCOCl}$ (e) $(\text{CH}_3\text{CH}_2)_2\text{O}$ (f) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CN}$
 (g) $(\text{CH}_3)_3\text{CCH}_2\text{CH}_2\text{COOH}$ (h)  (i) 

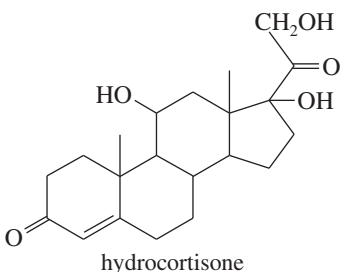
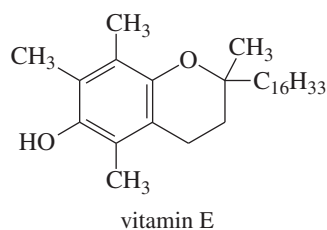
- (j)  (k)  (l) 
 (m)  (n)  (o) 

PROBLEM 2-27

Circle the functional groups in the following structures. State to which class (or classes) of compounds the structure belongs.

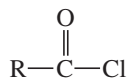
- (a) $\text{CH}_2=\text{CHCH}_2\text{COOCH}_3$ (b) CH_3OCH_3 (c) CH_3CHO
 (d) CH_3CONH_2 (e) CH_3NHCH_3 (f) RCOOH

- (g)  (h)  (i) 

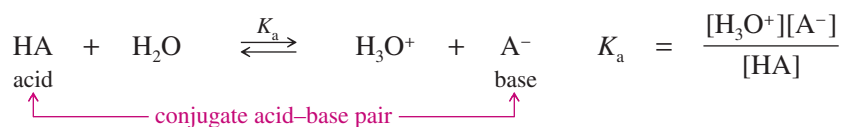
- (j)  hydrocortisone (k)  vitamin E

Essential Terms**acid chloride**

An acid derivative with a chlorine atom in place of the hydroxy group. (p. 135)

**acid-dissociation constant (K_a)**

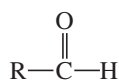
The equilibrium constant for the reaction of the acid with water to generate H_3O^+ . (p. 109)



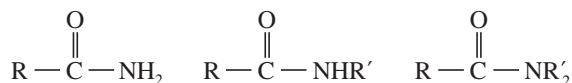
The negative logarithm of K_a is expressed as $\text{p}K_a$:

$$\text{p}K_a = -\log_{10} K_a$$

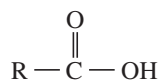
acids and bases	(p. 107)
(Arrhenius definitions)	acid: dissociates in water to give H_3O^+ base: dissociates in water to give OH^-
(Brønsted–Lowry definitions)	acid: proton donor base: proton acceptor
(Lewis definitions)	acid: electron-pair acceptor (electrophile) base: electron-pair donor (nucleophile)
alcohol	A compound that contains a hydroxy group bonded to a carbon atom; $\text{R}-\text{OH}$. (p. 133)
aldehyde	A carbonyl group with one alkyl group and one hydrogen. (p. 133)



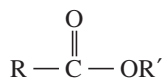
alkanes	Hydrocarbons containing only single bonds. (p. 129)
alkenes	Hydrocarbons containing one or more $\text{C}=\text{C}$ double bonds. (p. 131)
alkyl group	A hydrocarbon group with only single bonds; an alkane with one hydrogen removed, to allow bonding to another group; symbolized by R. (p. 130)
alkynes	Hydrocarbons containing one or more $\text{C}\equiv\text{C}$ triple bonds. (p. 131)
amide	An acid derivative that contains a nitrogen atom instead of the hydroxy group of the acid. (p. 135)



amine	An alkylated analog of ammonia; $\text{R}-\text{NH}_2$, R_2NH , or R_3N . (p. 135)
aromatic hydrocarbons	(arenes) Hydrocarbons containing a <i>benzene ring</i> , a six-membered ring with three double bonds. (p. 132)
bond dipole moment	A measure of the polarity of an individual bond in a molecule, defined as $\mu = (4.8 \times d \times \delta)$, where μ is the dipole moment in debyes (10^{-10} esu-Å), d is the bond length in angstroms, and δ is the effective amount of charge separated, in units of the electronic charge. (p. 94)
carbonyl group	The >C=O functional group, as in a ketone or an aldehyde. (p. 133)
carboxyl group	The $-\text{COOH}$ functional group, as in a carboxylic acid. (p. 134)
carboxylic acid	A compound that contains the carboxyl group, $-\text{COOH}$. (p. 134)



conjugate acid	The acid that results from protonation of a base. (p. 108)
conjugate base	The base that results from loss of a proton from an acid. (p. 108)
curved-arrow formalism	A method of drawing curved arrows to keep track of electron movement from nucleophile to electrophile (or within a molecule) during the course of a reaction. (p. 127)
cyano group	The $-\text{C}\equiv\text{N}$ functional group, as in a nitrile. (p. 136)
dipole–dipole forces	Attractive intermolecular forces resulting from the attraction of the positive and negative ends of the permanent dipole moments of polar molecules. (p. 98)
dipole moment (μ)	A measure of the polarity of a bond (or a molecule), proportional to the product of the charge separation times the bond length. (p. 94)
electrophile	An electron-pair acceptor (Lewis acid). (p. 126)
ester	An acid derivative with an alkyl group replacing the acid proton. (p. 135)



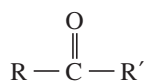
ether	A compound with an oxygen bonded between two alkyl (or aromatic) groups; $\text{R}-\text{O}-\text{R}'$. (p. 133)
-------	---

functional group	The reactive, nonalkane part of an organic molecule. (p. 130)
hydrocarbons	Compounds composed exclusively of carbon and hydrogen.
alkanes:	Hydrocarbons containing only single bonds. (p. 129)
alkenes:	Hydrocarbons containing one or more C=C double bonds. (p. 131)
alkynes:	Hydrocarbons containing one or more C≡C triple bonds. (p. 131)
cycloalkanes, cycloalkenes, cycloalkynes:	Alkanes, alkenes, and alkynes in the form of a ring. (p. 130)
aromatic hydrocarbons:	Hydrocarbons containing a <i>benzene ring</i> , a six-membered ring with three double bonds. (p. 132)



benzene

hydrogen bond	A particularly strong attraction between a nonbonding pair of electrons and an electrophilic O—H or N—H hydrogen. Hydrogen bonds have bond energies of about 20 kJ/mol (5 kcal/mol), compared with about 400 kJ/mol (about 100 kcal/mol) for typical C—H bonds. (p. 100)
hydrophilic	(“water-loving”) Polar substances or groups that readily dissolve in water; water-attractive. (p. 105)
hydrophobic	(“water-hating”) Nonpolar substances or groups that do not readily dissolve in water; water-repellant. (p. 104)
hydroxy group	(hydroxyl group) The —OH functional group, as in an alcohol. (p. 133)
isomers	Different compounds with the same molecular formula. (p. 81)
ketone	A carbonyl group bonded to two alkyl groups. (p. 133)



Lewis acid	An electron-pair acceptor (electrophile). (p. 126)
Lewis base	An electron-pair donor (nucleophile). (p. 126)
London dispersion forces	Intermolecular forces resulting from the attraction of correlated temporary dipole moments induced in adjacent molecules. (p. 99)
miscible	Soluble in all proportions. (p. 107)
molecular dipole moment	The vector sum of the bond dipole moments (and any nonbonding pairs of electrons) in a molecule; a measure of the polarity of a molecule. (p. 95)
nitrile	A compound containing a cyano group, —C≡N. (p. 136)
nucleophile	An electron-pair donor (Lewis base). (p. 126)
pH	A measure of the acidity of a solution, defined as the negative logarithm (base 10) of the H ₃ O ⁺ concentration: pH = −log ₁₀ [H ₃ O ⁺]. (p. 107)
van der Waals forces	The attractive forces between neutral molecules, including dipole–dipole forces and London dispersion forces. (p. 99)
dipole–dipole forces:	The forces between polar molecules resulting from the attraction of their permanent dipole moments.
London forces:	Intermolecular forces resulting from the attraction of correlated temporary dipole moments induced in adjacent molecules.

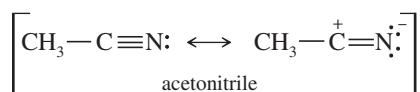
Essential Problem-Solving Skills in Chapter 2

Each skill is followed by problem numbers exemplifying that particular skill.

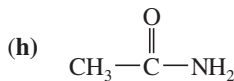
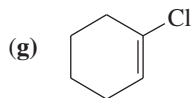
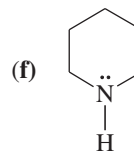
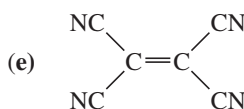
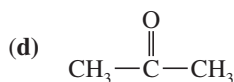
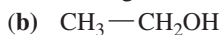
1	Identify polar and nonpolar molecules. Identify those molecules that engage in hydrogen bonding, either between themselves or with other hydrogen-bond donors.	Problems 2-28, 29, 30, 31, and 32
2	Predict general trends in the boiling points and solubilities of compounds based on their size, polarity, and ability to form hydrogen bonds.	Problems 2-31, 32, 33, 34, and 35
3	Predict relative acidities and basicities based on the structure, bonding, and resonance of conjugate acid–base pairs.	Problems 2-36, 37, 38, 40, 45, 46, 47, 48, and 54
4	Calculate and interpret values of K_a , K_b , pK_a , and pK_b . Use these values, and values for similar compounds, to predict products of acid–base reactions.	Problems 2-39, 41, 42, 43, 44, and 48
	<i>Problem-Solving Strategy: Predicting Acid–Base Equilibrium Positions</i>	Problems 2-39, 42, 43, and 49
5	Identify electrophiles (Lewis acids) and nucleophiles (Lewis bases), and write equations for Lewis acid–base reactions using curved arrows to show the flow of electrons.	Problems 2-50, 51, 52, and 53
6	Identify alkanes, alkenes, and aromatic hydrocarbons, and draw structural formulas for examples.	Problems 2-55, 56, and 57
7	Identify alcohols, ethers, aldehydes, ketones, carboxylic acids, and esters, and draw structural formulas for examples.	Problems 2-55, 56, and 57
8	Identify amines, amides, and nitriles, and draw structural formulas for examples.	Problems 2-55, 56, and 57

Study Problems

- 2-28** The $\text{C}\equiv\text{N}$ triple bond in acetonitrile has a dipole moment of about 3.6 D and a bond length of about 1.16 Å. Calculate the amount of charge separation in this bond. How important is the charge-separated resonance form in the structure of acetonitrile?

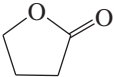
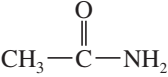


- 2-29** For each of the following compounds,
- draw the Lewis structure.
 - show how the bond dipole moments (and those of any nonbonding pairs of electrons) contribute to the molecular dipole moment.
 - estimate whether the compound will have a large, small, or zero dipole moment.

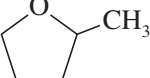
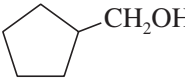
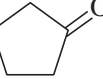
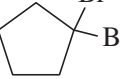


- 2-30** Ozone has a dipole moment of 0.53 D. Carbon dioxide has a dipole moment of zero, even though $\text{C}-\text{O}$ bonds are more polar than $\text{O}-\text{O}$ bonds. Explain this apparent contradiction.

2-31 Which of the following pure compounds can form hydrogen bonds? Which can form hydrogen bonds with water? Which ones do you expect to be soluble in water?

- (a) $(\text{CH}_3\text{CH}_2)_2\text{NH}$ (b) $(\text{CH}_3\text{CH}_2)_3\text{N}$ (c) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
 (d) $(\text{CH}_3\text{CH}_2\text{CH}_2)_2\text{O}$ (e) $\text{CH}_3(\text{CH}_2)_3\text{CH}_3$ (f) $\text{CH}_2=\text{CH}-\text{CH}_2\text{CH}_3$
 (g) CH_3COCH_3 (h) $\text{CH}_3\text{CH}_2\text{COOH}$ (i) $\text{CH}_3\text{CH}_2\text{CHO}$
 (j)  (k)  (l) 

2-32 Predict which member of each pair is more soluble in water. Explain your prediction.

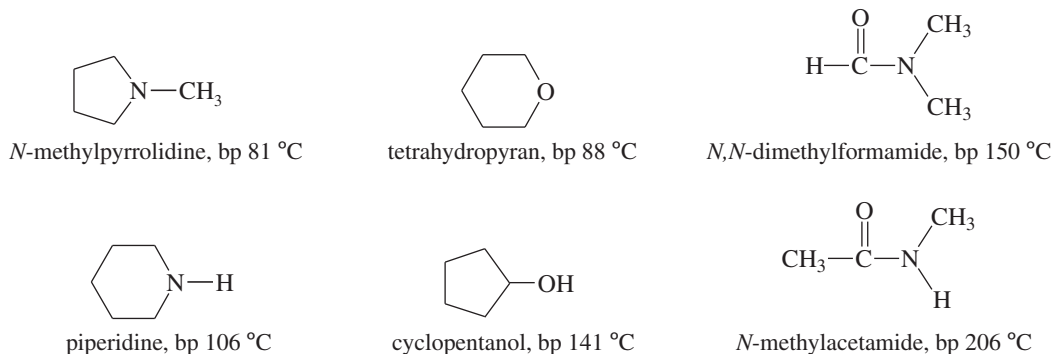
- (a)  or  (b)  or  (c)  or 
 (d)  or  (e)  or  (f)  or 

2-33 Dimethyl ether and ethanol are isomers. Their boiling points are very different, however. Explain why these two compounds have dramatically different boiling points.

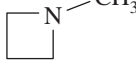
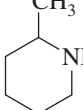
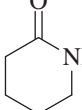


2-34 *N*-Methylpyrrolidine has a boiling point of 81°C , and piperidine has a boiling point of 106°C .

- (a) Explain the large difference (25°C) in boiling point for these two isomers.
 (b) Tetrahydropyran has a boiling point of 88°C , and cyclopentanol has a boiling point of 141°C . These two isomers have a boiling point difference of 53°C . Explain why the two oxygen-containing isomers have a much larger boiling point difference than the two amine isomers.
 (c) *N,N*-Dimethylformamide has a boiling point of 150°C , and *N*-methylacetamide has a boiling point of 206°C , for a difference of 56°C . Explain why these two nitrogen-containing isomers have a much larger boiling point difference than the two amine isomers. Also explain why these two amides have higher boiling points than any of the other four compounds shown (two amines, an ether, and an alcohol).



2-35 Predict which compound in each pair has the higher boiling point. Explain your prediction.

- (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OCH}_3$ or $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$ (b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ or $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
 (c) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ or $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{CH}_3$ (d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ or $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$
 (e)  or  (f)  or 

2-36 All of the following compounds can react as acids. Without using a table of acidities, rank them in order of increasing acidity. Explain your ranking.

- (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{SO}_3\text{H}$ (b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (c) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$
 (d) $\text{CH}_3\text{CH}_2\text{CHClCOOH}$ (e) $\text{ClCH}_2\text{CH}_2\text{CH}_2\text{COOH}$

(continued)

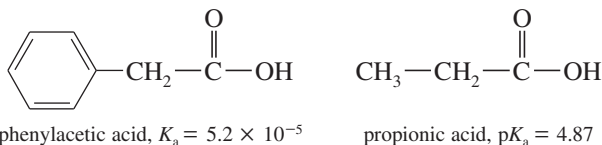
- 2-37 Rank the following species in order of increasing acidity. Explain your reasons for ordering them as you do.



- 2-38 Rank the following species in order of increasing basicity. Explain your reasons for ordering them as you do.



- 2-39 The K_a of phenylacetic acid is 5.2×10^{-5} , and the pK_a of propionic acid is 4.87.

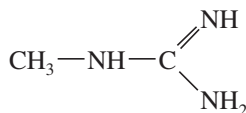


- (a) Calculate the pK_a of phenylacetic acid and the K_a of propionic acid.
 (b) Which of these is the stronger acid? Calculate how much stronger an acid it is.
 (c) Predict whether the following equilibrium will favor the reactants or the products.

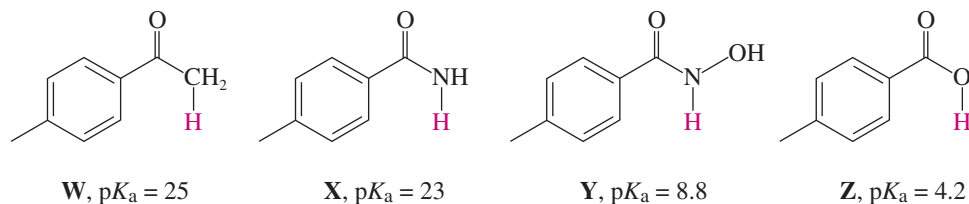


- *2-40 The following compound can become protonated on any of the three nitrogen atoms. One of these nitrogens is much more basic than the others, however.

- (a) Draw the important resonance forms of the products of protonation on each of the three nitrogen atoms.
 (b) Determine which nitrogen atom is the most basic.

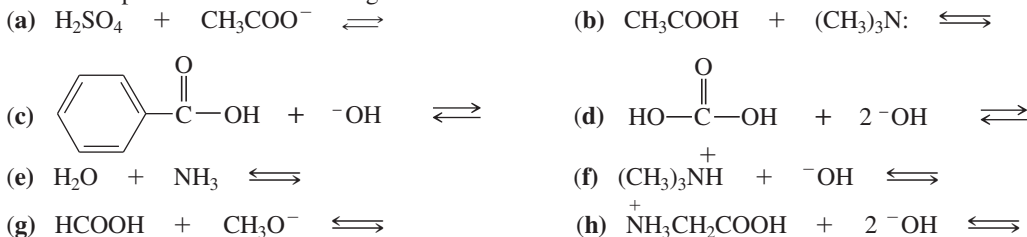


- 2-41 The following compounds are listed in increasing order of acidity. In each case, the most acidic proton is shown in red.

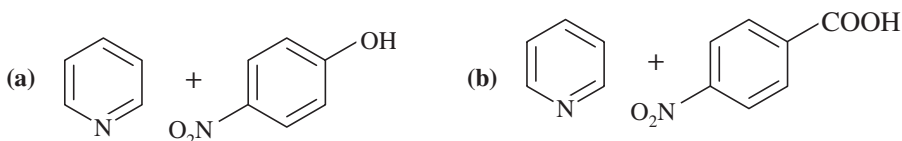


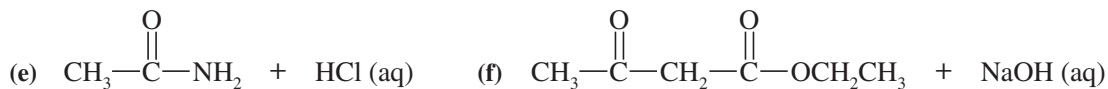
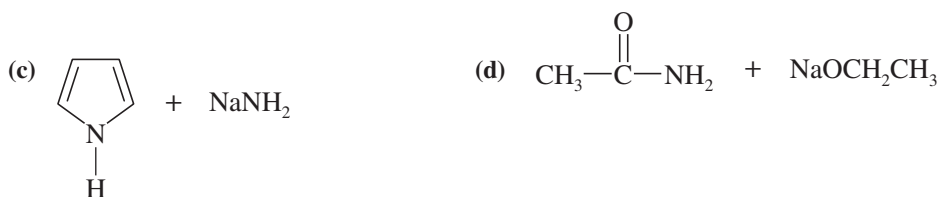
- (a) Show the structure of the conjugate base of each acid, including any resonance forms.
 (b) Explain why X is a stronger acid than W.
 (c) Explain why Y is a stronger acid than X.
 (d) Explain why Z is a stronger acid than Y.

- 2-42 Predict the products of the following acid–base reactions.

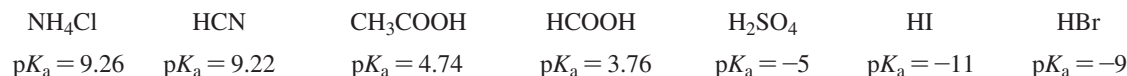


- 2-43 Consider the following proposed Brønsted–Lowry acid–base reactions. In each case, draw the products of a transfer of the most acidic proton on the acid to the most basic site on the base. Use Appendix 4 to find or estimate the pK_a values for the acids and the pK_b values for the bases. Then determine which side of the reaction is favored, either reactants or products.

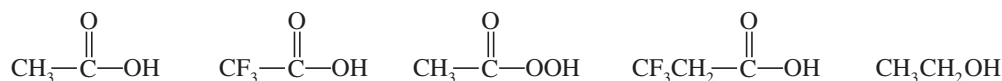




2-44 Compare the relative acidity of 1-molar aqueous solutions of the following acids.



2-45 The following compounds can all react as acids.



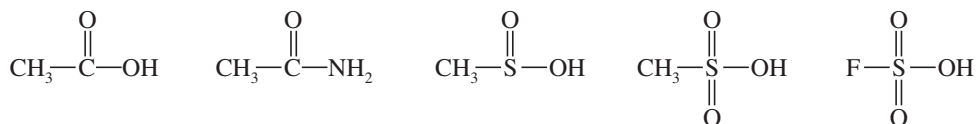
- For each compound, show its conjugate base. Show any resonance forms if applicable.
- Rank the conjugate bases in the order you would predict, from most stable to least stable.
- Rank the original compounds in order, from strongest acid to weakest acid.

2-46 The following compounds can all react as bases.



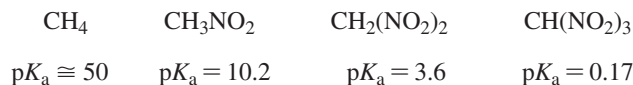
- For each compound, show its conjugate acid. Show any resonance forms if applicable.
- Rank the conjugate acids in the order you would predict, from most stable to least stable.
- Rank the original compounds in order, from strongest base to weakest base.

2-47 The following compounds can all react as acids.



- For each compound, show its conjugate base. Show any resonance forms if applicable.
- Rank the conjugate bases in the order you would predict, from most stable to least stable.
- Rank the original compounds in order, from strongest acid to weakest acid.

2-48 Consider the following compounds that vary from nearly nonacidic to strongly acidic. Draw the conjugate bases of these compounds, and explain why the acidity increases so dramatically with substitution by nitro groups.



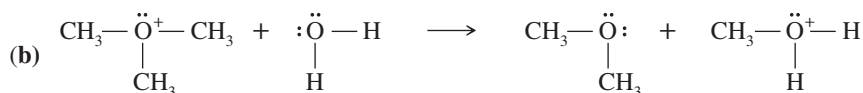
2-49 Ethyllithium (CH₃CH₂Li) is often used as a base in organic reactions.

- (a) Predict the products of the following acid–base reaction.

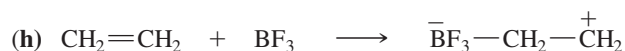
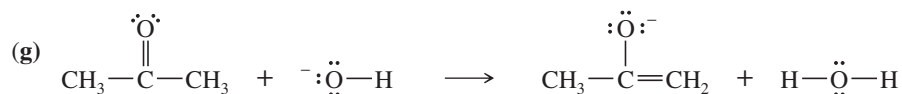
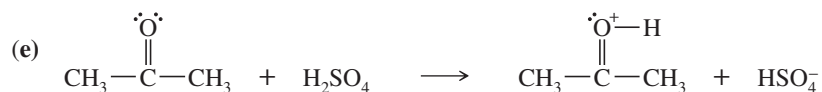
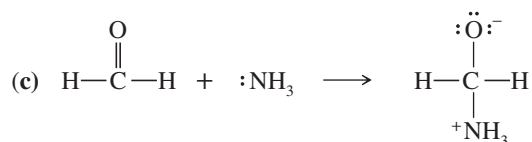


- (b) What is the conjugate acid of CH₃CH₂Li? Would you expect CH₃CH₂Li to be a strong base or a weak base?

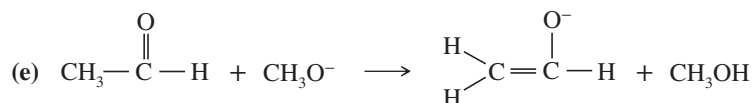
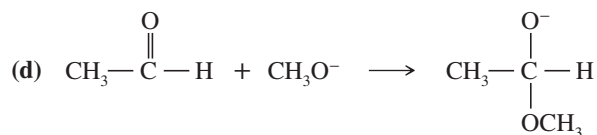
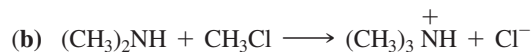
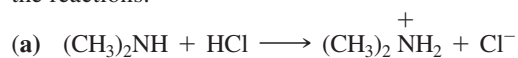
2-50 Label the reactants in these acid–base reactions as Lewis acids (electrophiles) or Lewis bases (nucleophiles). Use curved arrows to show the movement of electron pairs in the reactions.



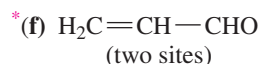
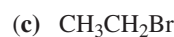
(continued)



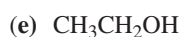
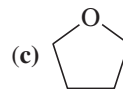
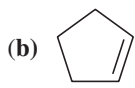
2-51 In each reaction, label the reactants as Lewis acids (electrophiles) or Lewis bases (nucleophiles). Use curved arrows to show the movement of electron pairs in the reactions. Draw any nonbonding electrons to show how they participate in the reactions.



2-52 Each of these compounds can react as an electrophile. In each case, use curved arrows to show how the electrophile would react with the strong nucleophile sodium ethoxide, $\text{Na}^+ ^-\text{OCH}_2\text{CH}_3$.



2-53 Each of these compounds can react as a nucleophile. In each case, use curved arrows to show how the nucleophile would react with the strong electrophile BF_3 .

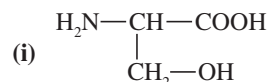
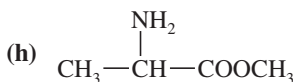
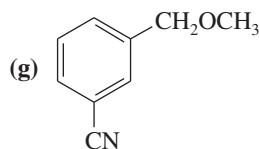
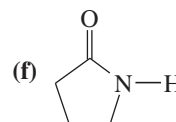
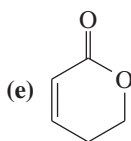
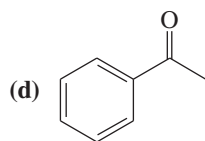
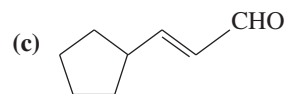
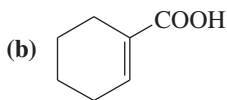
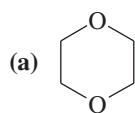


- 2-54** The pK_a of ascorbic acid (vitamin C, page 93) is 4.17, showing that it is slightly more acidic than acetic acid (CH_3COOH , pK_a 4.74).
- Show the four different conjugate bases that would be formed by deprotonation of the four different OH groups in ascorbic acid.
 - Compare the stabilities of these four conjugate bases, and predict which OH group of ascorbic acid is the most acidic.
 - Compare the most stable conjugate base of ascorbic acid with the conjugate base of acetic acid, and suggest why these two compounds have similar acidities, even though ascorbic acid lacks the carboxylic acid (COOH) group.

2-55 Give a definition and an example for each class of organic compounds.

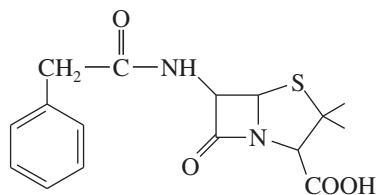
- | | | |
|--------------|--------------------------|---------------------|
| (a) alkane | (b) alkene | (c) alkyne |
| (d) alcohol | (e) ether | (f) ketone |
| (g) aldehyde | (h) aromatic hydrocarbon | (i) carboxylic acid |
| (j) ester | (k) amine | (l) amide |
| (m) nitrile | | |

2-56 Circle the functional groups in the following structures. State to which class (or classes) of compounds the structure belongs.

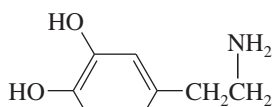


2-57 Many naturally occurring compounds contain more than one functional group. Identify the functional groups in the following compounds:

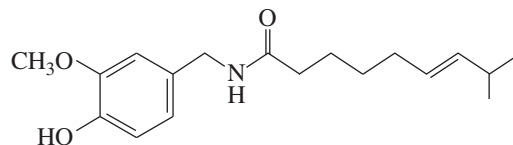
- Penicillin G is a naturally occurring antibiotic.
- Dopamine is the neurotransmitter that is deficient in Parkinson's disease.
- Capsaicin gives the fiery taste to chili peppers.
- Thyroxine is the principal thyroid hormone.
- Testosterone is a male sex hormone.



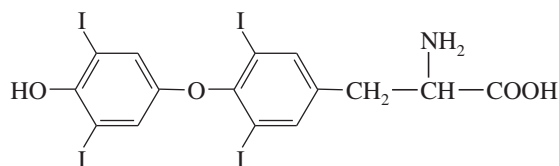
penicillin G



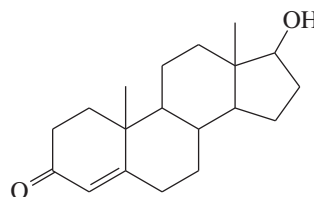
dopamine



capsaicin



thyroxine



testosterone